

Expansion-assisted selective plane illumination microscopy for nanoscale imaging of centimeter-scale tissues

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Abstract

Recent advances in tissue processing, labeling, and fluorescence microscopy are providing unprecedented views of the structure of cells and tissues at sub-diffraction resolutions and near single molecule sensitivity, driving discoveries in diverse fields of biology, including neuroscience. Biological tissue is organized over scales of nanometers to centimeters. Harnessing molecular imaging across intact, three-dimensional samples on this scale requires new types of microscopes with larger fields of view and working distance, as well as higher throughput. We present a new expansion-assisted selective plane illumination microscope (ExA-SPIM) with aberration-free $1 \times 1 \times 3 \mu\text{m}$ optical resolution over a large field of view ($10.6 \times 8.0 \text{ mm}^2$) and working distance (35 mm) at speeds up to 946 megavoxels/sec. Combined with new tissue clearing and expansion methods, the microscope allows imaging centimeter-scale samples with $250 \times 250 \times 750 \text{ nm}$ optical resolution (4× expansion), including entire mouse brains, with high contrast and without sectioning. We illustrate ExA-SPIM by reconstructing individual neurons across the mouse brain, imaging cortico-spinal neurons in the macaque motor cortex, and visualizing axons in human white matter.

eLife assessment

The ExA-SPIM methodology developed will be **important** to the field of light sheet microscopy as the new technology provides an impressive field of view making it possible to image the entire expanded mouse brain at cellular and subcellular resolution. The authors provide **solid** evidence that mostly supports the conclusions.

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Introduction

Biological tissue is organized over scales of nanometers to centimeters. Understanding individual cells and multi-cellular organization requires probing the architecture of tissue over these spatial scales simultaneously. However, standard microscope objectives with high resolutions have limited working distances (<1 mm) and fields of view (<1 mm). Imaging large tissue volumes at high resolutions therefore requires physical sectioning and extensive tiling. Physical sectioning distorts the imaged tissue and focal planes have optical distortions at the edges of the field of view (Zhang & Gross 2019). These factors complicate stitching across sections and tile boundaries at sub-micrometer resolutions which in turn increases the complexity and cost of downstream image analysis.

Because light aberration and scattering limits high-resolution microscopy in tissue, including two-photon microscopy, to depths of hundreds of micrometers, sectioning and tiling have traditionally been viewed as necessary to image large specimens (Portera-Cailliau et al. 2005; Tsai et al. 2009; Oh et al. 2014; Economo et al. 2016). However, advances in histological methods, including clearing (Richardson & Lichtman 2015; Spalteholz 1914; Dodt et al. 2007; Tsai et al. 2009; Hama et al. 2011; Becker et al. 2012; Ertürk et al. 2012; Chung & Deisseroth 2013; Ke et al. 2013; Susaki et al. 2014; Tainaka et al. 2014; Yang et al. 2014; Renier et al. 2014; Hou et al. 2015; Costantini et al. 2015; AU - Chi et al. 2018) and expansion for microscopy (ExM) (Chen et al. 2015; Chen et al. 2016; Chozinski et al. 2016; Ku et al. 2016), now promise diffraction-limited imaging deep in tissue. Avoiding sectioning and reducing tiling requires overcoming the “volumetric” imaging barrier of microscopy (i.e., the maximum volume that may be imaged) (Figure 1a). Expansion up to $20\times$ has been demonstrated (Chang et al. 2017), which provides access to molecular spatial scales (10's of nm), approaching those of cryo electron microscopy (a few nm) (Cheng et al. 2015; Fernandez-Leiro & Scheres 2016; Nogales & Scheres 2015). These ExM methods produce large, fragile three-dimensional samples, further emphasizing the need for overcoming the volumetric imaging barrier.

We combine a new microscope and methods for tissue clearing and expansion, which we jointly refer to as Expansion-Assisted Selective Plane Illumination Microscopy (ExA-SPIM). Unlike expansion lattice light-sheet microscopy, (G. Gao et al. 2023) which is limited to small tissue volumes $\ll 1\text{ mm}^3$, ExA-SPIM works with expansion of large centimeter-scale tissue volumes, such as the mouse brain. Leveraging optics and detectors developed for the electronics metrology industry, ExA-SPIM has a field of view $\sim 100\times$ larger (13.3 mm diameter) and a working distance $\sim 10\times$ larger (35 mm) compared to objectives typically used for biological microscopy, while retaining a relatively high numerical aperture (NA) = 0.305. When combined with $3\times$ expansion the system achieves an effective optical resolution of $\sim 0.33\text{ }\mu\text{m}$ laterally, and $\sim 1\text{ }\mu\text{m}$ axially, at imaging speeds of up to 946 megavoxels/sec. Imaging with diffraction-limited resolution throughout centimeter-scale specimens requires tissue samples with small index of refraction variations (Weiss et al. 2021). To achieve this goal, we developed a novel expansion method for centimeter-scale tissue samples. Tailoring the expansion factor allows fine tuning the effective resolution for specific tissue types and scientific questions.

We apply ExA-SPIM to imaging and reconstructing mammalian neurons. Axonal arbors of individual neurons are complex, branched structures that transmit electrical impulses over distances of centimeters, yet axons can be thinner than 100 nm (Economo et al. 2016; Winnubst et al. 2019). Axons contain numerous varicosities that make synapses with other neurons. Tracing the axonal arbors of single neurons is critical to define how signals are routed within the brain and is also necessary for classifying diverse neuron types into distinct types (Economo et al. 2016; Wang et al. 2021; Winnubst et al. 2019; Xu et al. 2021; Zhang et al. 2021). Large-scale projects based on single cell transcriptomics in the mouse and human brain have revealed a

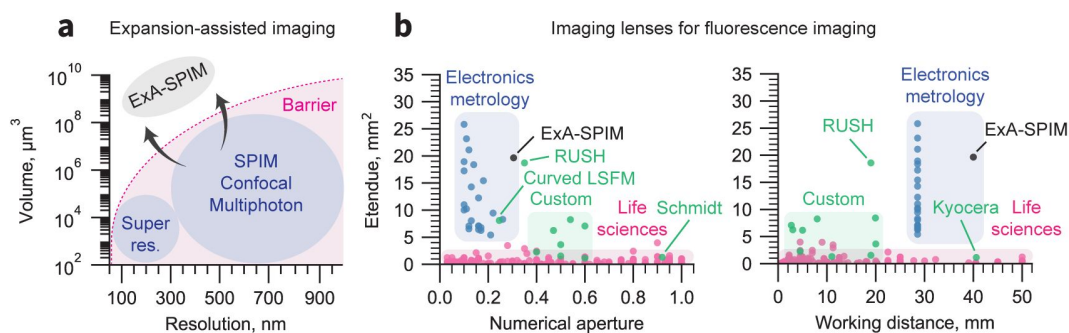


Figure 1

Breaking the volumetric imaging barrier.

(a) Current fluorescence microscopy approaches are bounded by a volumetric imaging barrier (thick pink line; inspired by (Daetwyler & Fiolka 2023 [link](#))). Resolution is limited by the diffraction limit. The accessible imaging volume is limited by specifications of life sciences microscope objectives. The former can be surpassed by using tissue expansion, and the latter can be overcome by using highly engineered lenses from the electronics metrology industry. (b) The etendue (G) of 90% of life sciences objectives (magenta) are $< 1 \text{ mm}^2$ (Yueqian & Herbert 2019 [link](#)), apart from several custom lenses (green). In contrast, lenses developed for electronics metrology can have $G > 10 \text{ mm}^2$. The lens used in the ExA-SPIM system provides a field of view of 16.8 mm^2 with $\text{NA} = 0.305$ ($G = 19.65 \text{ mm}^2$). This etendue is comparable to the custom RUSH objective (Fan et al. 2019 [link](#)), but with twice the working distance and correction for liquid media. The RUSH, Schmidt (Voigt et al. 2024 [link](#)), Kyocera^a, and large etendue curved focal plane (Tang et al. 2024 [link](#)) lenses that lie outside of their colored zone are highlighted.

a. <https://www.ksoc.co.jp/en/seihin/immersion-objective/immersion-objective.html> [link](#)

great diversity of neuron types (Hodge et al. 2019 [↗](#); Tasic et al. 2018 [↗](#); Zizhen Yao et al. 2023 [↗](#)). However, the throughput of neuronal reconstructions has remained too low for single neuron reconstructions on a comparable scale, even in mice. Throughput is limited in part by speed and image quality of existing microscopy methods. We demonstrate that ExA-SPIM provides high-resolution fluorescence microscopy over teravoxel image volumes with minimal distortions, and thereby enables brain-wide imaging with high contrast, resolution, and speed.

Results

We first describe the microscope optics and how they overcome specific requirements for multi-scale tissue imaging. We then outline a new histological method for clearing and expanding centimeter-scale specimens, although details are relegated to extensive protocols (Ouellette et al. 2023 [↗](#), [dx.doi.org/10.17504/protocols.io.n92ldpwjxl5b/v1](https://doi.org/10.17504/protocols.io.n92ldpwjxl5b/v1) [↗](#)). Finally, we illustrate ExA-SPIM performance for imaging neurons in whole mouse brains and large samples of non-human primate and human cortex.

ExA-SPIM microscope

The ideal fluorescence microscope for large-scale tissue imaging would provide: (1) nanoscale resolution, (2) over centimeter-scale volumes, (3) with minimal tiling and sectioning, (4) high isotropy in resolution and contrast, and (5) fast imaging speed. Multiple imaging systems have been developed within this space, each with unique advantages but inevitable trade-offs (Huisken et al. 2004 [↗](#); Dodt et al. 2007 [↗](#); Wu et al. 2013 [↗](#); Kumar et al. 2014 [↗](#); Tomer et al. 2014 [↗](#); Economo et al. 2016 [↗](#); Narasimhan et al. 2017 [↗](#); Power & Huisken 2017 [↗](#); Migliori et al. 2018 [↗](#); Chakraborty et al. 2019 [↗](#); Voigt et al. 2019 [↗](#); Voleti et al. 2019 [↗](#); Y. Chen et al. 2020 [↗](#); D. Chen et al. 2020 [↗](#); Glaser et al. 2022 [↗](#); Wang et al. 2021 [↗](#); Xu et al. 2021 [↗](#); Zhang et al. 2021 [↗](#); Xiaoli Qi et al. 2023 [↗](#); Vladimirov et al. 2024 [↗](#)).

The choice of microscope objective is a critical design choice. Microscope objectives impose trade-offs between the smallest objects that can be resolved (i.e., the resolution), how much of the specimen can be observed at once (i.e., the field of view), and how thick of a specimen may be imaged (i.e., the working distance). These trade-offs are not based on physical law, but reflect limitations of optical design, engineering, and lens manufacturing.

The trade-off between resolution and field of view is related to the etendue (G), which is proportional to the number of resolution elements of an optical system. The etendue is a quadratic function of the lens field of view (FOV) and numerical aperture (NA).

$$G = \frac{\pi}{4} (FOV \times NA)^2 \quad (1)$$

Similarly, for a given NA, the required lens diameter increases proportionally with the required working distance (WD). The ideal lens for large-scale volumetric microscopy would provide large G and WD values.

90% of commercially available objectives for biological microscopy have $G < 1 \text{ mm}^2$ (Yueqian & Herbert 2019 [↗](#)), and the vast majority of commercially available lenses have lens diameters $< 2 \text{ cm}$. Several attempts have been made to develop custom lenses for specific applications. The ‘Mesolens’ provides an etendue of $G = 6.25 \text{ mm}^2$, with a FOV = 6 mm, NA = 0.47, and WD = 3 mm (McConnell et al. 2016 [↗](#)), but difficulties with manufacturing has limited adoption (McConnell 2020 [↗](#)). An objective with an etendue of $G = 7.07 \text{ mm}^2$, FOV = 5 mm, NA = 0.6, and WD = 2.7 mm has been developed for two-photon microscopy *in vivo* (Sofroniew et al. 2016 [↗](#)). However, this lens is customized for infrared illumination and exhibits significant field curvature. Additional custom lenses have subsequently been developed for two-photon microscopy, although none of

these lenses are suitable for large-scale fluorescence microscopy (Keisuke Ota et al. 2020 [↗](#); Rumyantsev et al. 2020 [↗](#); Yu et al. 2021 [↗](#)). A custom lens with a large etendue of $G = 18.86 \text{ mm}^2$, FOV = 14 mm, NA = 0.35, and WD = 19 mm has been developed for the real-time ultra-large-scale high-resolution (RUSH) microscope (Fan et al. 2019 [↗](#)). Despite the large etendue, the lens is not commercially available and not designed for immersion, which is critical for imaging cleared or expanded tissues. Custom lenses such as the Schmidt (Voigt et al. 2024 [↗](#)), Cousa (Yu et al. 2024 [↗](#)), and Kyocera (<https://www.ksoc.co.jp/en/seihin/immersion-objective/immersion-objective.html> [↗](#)) lenses offer long working distances but small etendues.

Rather than designing a highly customized lens from scratch, we leveraged engineering investments that have been made in a different domain. High-resolution, high-speed imaging is widely used within the machine vision and metrology industry, where optical microscopes are used to map defects in semiconductors and other electronic devices. As the physical size of electronic components (e.g., pixels on flat panel displays) have become smaller, lenses for this domain have been designed with increasing NA, accessing biologically relevant resolutions ($<1 \mu\text{m}$). For a given NA these lenses have remarkably large field of views (and thus high etendues), low field curvature, minimal distortion, and chromatic correction throughout the visible wavelengths. A comparison of the etendue as a function of NA and working distance for these lenses, and commercial and custom life sciences lenses is summarized in (Figure 1b [↗](#)). A more detailed summary of many electronics metrology technologies (<https://schneiderkreuznach.com/en/industrial-optics/lenses/line-scan-lenses/diamond> [↗](#); <https://www.excelitas.com/product/dfine-hr-24128-lens> [↗](#); <https://navitar.com/products/imaging-optics/singleshot/> [↗](#); <https://www.nikon.com/business/industrial-lenses/> [↗](#)) is provided as **Supplementary Data**. Despite their superb specifications, both the lenses and camera sensors are readily available, are manufactured in large quantities, and are cost effective. We investigated the performance of metrology optics and cameras for imaging biological tissues and adapted a particular set of machine vision optics for biological microscopy.

The detection path of the system uses a lens (https://vision.viewworks.com/en/lens/veo_jm/VEO_JM_DIAMOND_5.0X_F1.3 [↗](#)) with 5.0× magnification, with diffraction-limited imaging at NA = 0.305, and a 16.8 mm field of view ($G = 19.65 \text{ mm}^2$) (VEO_JM DIAMOND 5.0×/F1.3, Schneider-Kreuznach, Germany) (Figure 2a [↗](#)). A 35 mm thick glass beam splitter, normally used for co-axial illumination, is replaced with an optically equivalent thickness of liquid media. This enables spherical aberration free imaging through 35 - 40 mm of liquid media (tunable for refractive indices between 1.33 - 1.56), including expanded hydrogels and cleared tissues. The lens is paired with a large-format CMOS sensor (https://vision.viewworks.com/en/camera/area_scan/VP_series [↗](#)) (VP-151MX, Viewworks Korea), based on the Sony IMX411 sensor (https://www.sony-semicon.com/files/62/pdf/p-13_IMX411ALR_AQR_Flyer.pdf [↗](#)), with a single-sided rolling shutter, 151 megapixels, 14192 (H) × 10640 (V), and a pitch of 3.76 μm . The camera captures a field of view of 10.6×8.0 mm (13.3 mm diagonal), capable of covering an entire 3× expanded mouse brain in only 15 tiles (Figure 2b [↗](#)). A conventional SPIM system would require 400+ tiles to image an entire cleared brain at equivalent resolutions (Figure S1 [↗](#)). A comparison of the selected lens and camera sensor with a state-of-the-art cleared tissue objective lens (Nikon 20× GLYC) and sCMOS camera (Hamamatsu Orca BT-Fusion) are shown in Figure S2 [↗](#).

To improve axial resolution, our system synchronizes an axially-swept light-sheet with the rolling shutter of the Sony IMX411 sensor (Dean et al. 2015 [↗](#)). The CMOS sensor parallelizes readout across 14192 pixels per row, thereby achieving equivalent or greater pixel rates than typical scientific CMOS (sCMOS) sensors, even with an increased time per line and a relatively low frame rate ($<6.4 \text{ Hz}$). As a result, the ExA-SPIM microscope operates at a higher imaging speed with comparable signal-to-noise ratio (SNR) to sCMOS-based systems (Figure 2e [↗](#)). The low frame rate of the sensor facilitates accurate axially swept imaging using generic scanning hardware at an imaging speed of 946 megavoxels/sec. The excitation lens (1-290419, Navitar) is infinity-corrected (to enable axially swept excitation) and provides diffraction-limited resolution at NA = 0.133 over a

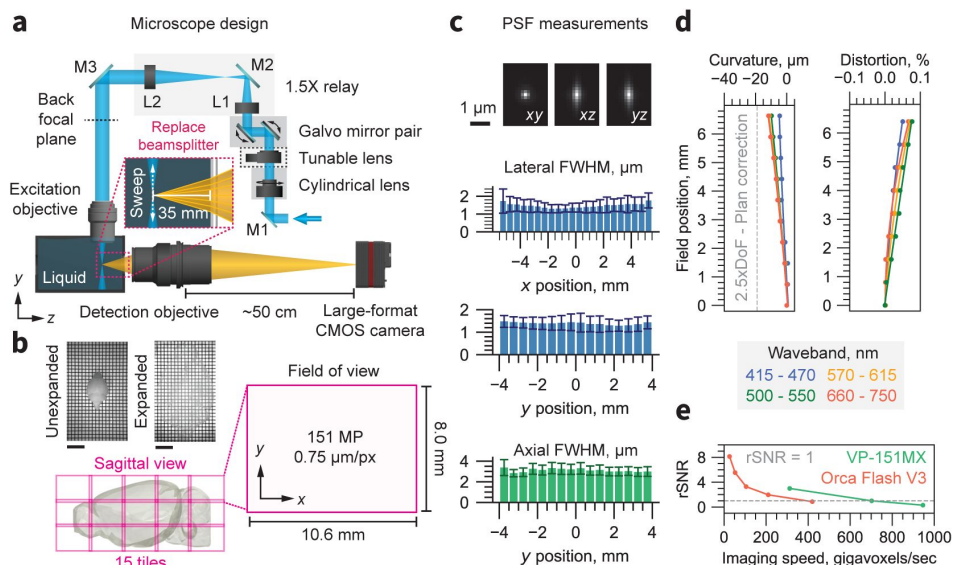


Figure 2

Microscope overview.

(a) Schematic of the ExA-SPIM system. Light enters the system from the laser combiner and is reflected by mirror M1. A cylindrical lens focuses the light in one dimension onto the surface of a tunable lens, which is magnified onto the back focal plane of the excitation through a 1.5× relay consisting of lenses L1 and L2 and mirrors M2 and M3. The excitation objective is oriented vertically and dipped into a liquid immersion chamber. The tunable lens is conjugated to the back focal plane of the excitation objective to enable axial sweeping. A pair of galvo mirrors are used in tandem to translate the position of the light sheet in *z* (along the optical axis of the detection objective). The detection objective is oriented horizontally. A beam splitter is removed from the lens and replaced with approximately 35 mm of water. A large-format CMOS camera captures images from the detection lens, at a back focusing distance of 50 cm. **(b)** The field of view of the system is 10.6×8.0 mm (13.3 mm diagonal), which is digitized by the camera into a 151-megapixel (MP) image with 0.75 μm/px sampling. Although the optical resolution of the detection lens is ~1.0 μm, the sampling limited resolution based on the Nyquist criterion is ~1.5 μm. The large field of view dramatically reduces the need for tiling. For example, a 3× expanded mouse brain can be captured in only 15 tiles. **(c)** The PSF for 561 nm excitation is shown in the *xy*, *xz*, and *yz* planes. The mean and standard deviation of the lateral and axial full-width half-maximum are shown as a function of *x* and *y* position across the full field of view. **(d)** The field curvature and distortion of the system as a function of field position is shown for different wavebands. The field curvature is <2.5× the depth of field (DoF) for all wavebands. This performance is better than “Plan” specified life sciences objectives (Yuegian & Herbert 2019). **(e)** The relative signal-to-noise-ratio (rSNR) of the VP-151MX CMOS camera and an Orca Flash V3 sCMOS camera as a function of imaging speed. The VP-151MX camera provides equivalent SNR at nearly twice the imaging speed.

16 mm field of view. When fitted with a custom dipping cap, the excitation lens provides a working distance of 53 mm in water. The beam shaping in the excitation path is configured to deliver a light sheet with NA = 0.10 and width of 12.5 mm (full-width half-maximum).

Based on these lenses, and the travel limits of a mechanical motorized stage, the ExA-SPIM is capable of imaging a $200 \times 52 \times 35 \text{ mm}^3$ volume, with $\sim 1 \text{ }\mu\text{m}$ lateral and $\sim 3 \text{ }\mu\text{m}$ axial native optical resolution, and minimal field curvature [quantified using the methodology described in (Vladimirov et al. 2024)] and distortion (**Figure 2c-d**). We leveraged this large volume to image intact, expanded tissues. The expansion factor allows tuning the effective resolution for specific biological applications.

With $3\times$ tissue expansion, the system can image a native tissue volume of $67 \times 17 \times 12 \text{ mm}^3$ with an effective optical resolution of $\sim 0.33 \text{ }\mu\text{m}$ laterally and $\sim 1 \text{ }\mu\text{m}$ axially. This amounts to >100 teravoxels, which can be captured without the need for physical sectioning and with minimal tiling. A summary plot comparing the focal volume, isotropy, and imaging speed of our new ExA-SPIM system to other large-scale volumetric imaging systems is shown in **Figure S3**. Individual data points are listed in **Table S2**.

The large-scale and high-speed imaging made possible by the ExA-SPIM system required new software for controlling the microscope, as well as downstream computational pipelines for compressing and handling the resulting datasets. We developed custom acquisition software capable of interfacing with the VP-151MX camera driver (<https://github.com/acquire-project>), which streams imaging data directly to next-generation file formats (Beati et al. 2020; Clack et al. 2023; Moore et al. 2021). Using a combination of high-speed networking, fast on-premises storage (<https://vastdata.com/>), and real-time compression, our current imaging pipeline enables a throughput to cloud storage of $>100 \text{ TB}$ per day, using commodity hardware. This pipeline is actively being developed, with ongoing efforts focused on standardizing every processing step around OME-Zarr (Moore et al. 2023).

Whole-brain expansion

Tissue expansion for microscopy (ExM) enables imaging specimens at effective resolutions well below the diffraction limit of light microscopes (Chang et al. 2017; Chen et al. 2015; Tillberg & Chen 2019). Moreover, ExM can produce optically clear specimens with low fluorescence background. Multiple ExM variations have emerged, driven by specific biological questions. These include engineered hydrogel chemistry for post expansion molecular interrogation of proteins or RNA (Asano et al. 2018; Cho & Chang 2022; Tillberg et al. 2016; Yi Cui et al. 2022), formulations that provide gel stiffness (R. Chen et al. 2021) or tunable expansion up to $>10\times$ (Damstra et al. 2022; Klimas et al. 2023; Sarkar et al. 2022; Truckenbrodt et al. 2018). Most of these protocols have been developed for specimens that are at most 100's of micrometers thick. We developed ExM methods for centimeter-scale tissue samples, including entire mouse brains. A key requirement for ExA-SPIM is optical clearing so that the entire volume can be imaged with diffraction-limited resolution without sectioning. Index of refraction inhomogeneities in heavily myelinated fiber tracts pose particular challenges. Clearing was achieved by stringent dehydration and delipidation prior to gelation and expansion. We systematically evaluated dehydration agents, including methanol, ethanol, and tetrahydrofuran (THF), followed by delipidation with commonly used protocols on 1 mm thick brain slices. Slices were expanded and examined for clarity under a macroscope. Dehydration using THF was followed by two sequential delipidation steps. First DISCO type clearing (AU - Chi et al. 2018; Renier et al. 2014) followed by aqueous delipidation ((Chen & Svoboda 2020), [dx.doi.org/10.17504/protocols.io.zndf5a6](https://doi.org/10.17504/protocols.io.zndf5a6)) rendered the samples extremely transparent (**Figure 3a**) ((Ouellette et al. 2023), [dx.doi.org/10.17504/protocols.io.n92dpwjl5b/v1](https://doi.org/10.17504/protocols.io.n92dpwjl5b/v1)). In addition to clearing specimens, delipidation facilitates immunolabeling brain samples prior to gelation and expansion (**Figure 3**). Signal amplification facilitates high contrast imaging of small structures, especially since the increase in volume upon expansion dilutes the concentration of the fluorophore. For gelation we used VA-044

as the initiator, instead of the more commonly used APS/TEMED (Tillberg & Chen 2019 [DOI](#)). VA-044 initiates free radicals at a temperature-dependent rate. At low temperatures (4°C) the gelling reagents are allowed to diffuse to the center of thick samples, followed by higher temperatures (37°C) to trigger uniform polymerization. Our protocol provides nearly isotropic expansion of the whole sample (including internal brain structures) (**Figure 3** [DOI](#)). Although developed for the mouse brain, we have successfully used this protocol for other large specimens, such as a 1×1×1.5 cm piece of macaque motor cortex (**Figure 5** [DOI](#)), as well as a 1×1×0.01 cm section of human visual cortex (**Figure 6** [DOI](#)) ((Ouellette et al. 2023 [DOI](#)), [dx.doi.org/10.17504/protocols.io.n92ldpwjx/5b/v1](https://doi.org/10.17504/protocols.io.n92ldpwjx/5b/v1) [DOI](#)).

Imaging single neurons across entire mouse brains

We cleared and expanded entire mouse brains and imaged individual neurons, resolving their dense axonal projections (**Figure 3** [DOI](#)). Tracking the axons of individual neurons is a challenging problem—axons collaterals can be very thin (<100 nm) and traverse large distances (centimeters), spanning vast areas of the brain. This requires high-resolution, high-contrast imaging of the entire brain without loss of data. Current best-in-class approaches (Economo et al. 2016 [DOI](#); Gong et al. 2013 [DOI](#); Gong et al. 2016 [DOI](#); Winnubst et al. 2019 [DOI](#)) require physical sectioning and extensive tiling, which complicates downstream data processing. In addition, imaging spans multiple days, which increases the chance for experimental failure and data loss. Finally, the resolution of these imaging methods is highly anisotropic (typically <1:6), which compromises the ability to perform unambiguous axon tracing.

We expressed GFP in a sparse subset of neurons in an enhancer virus mouse to label subcortical projection neurons. Brains were expanded (3×) and imaged in 15 tiles in as little as 24 hours. Because of the lack of physical tissue slicing, minimal tiling, and excellent performance of the microscope, individual tile volumes are nearly perfectly registered based on stage coordinates alone, as evident by the close alignment of axons across tile boundaries **Supplementary Videos 1-2**). Dense axonal projections and varicosities are clearly visible (**Figure 3c-h** [DOI](#)) and long-range axons can be tracked across the brain **Supplementary Videos 3-6**). The achromatic performance of the ExA-SPIM optics also enables multiple fluorescence channels to be acquired from mouse brains (e.g. GFP and tdTomato) (**Figure 4** [DOI](#)).

Imaging cortico-spinal tract neuron in macaque motor cortex

We next established that our methods are applicable to larger brains. With minor adaptations, we applied our clearing and expansion protocol to a 1×1×1.5 cm block of pigtail macaque brain from the hand-wrist and trunk regions of primary motor cortex. Due to its strong myelination, this region of the primate brain is particularly difficult to clear and image (Van Essen et al. 2019 [DOI](#)). The specimen contained cortico-spinal neurons expressing a fluorescent protein (tdTomato) driven by retrograde AAV injected into the intermediate and ventral laminae. ExA-SPIM image volumes revealed brightly labeled cortico-spinal neurons (**Figure 5a-b** [DOI](#) and **Supplementary Videos 7-9**). Individual neurons, their dendritic arbors, and extensive dendritic spines as well as the descending axon and collaterals are clearly discernible (**Figure 5c-f** [DOI](#) and **Supplementary Video 10**). Additionally, even in down sampled data (~1 μm effective voxel size), we can follow axonal pathways. This raises the possibility that with a small number of slices (~6×1 cm thick slabs) and reduced imaging resolution (e.g., ~1×1×2.5 μm), a modified inverted (Wu et al. 2013 [DOI](#); Kumar et al. 2014 [DOI](#)) or open-top (McGorty et al. 2015 [DOI](#); Glaser et al. 2017 [DOI](#); Glaser et al. 2019 [DOI](#); Barner et al. 2020 [DOI](#); Glaser et al. 2022 [DOI](#)) ExA-SPIM design (**Figure S4** [DOI](#)) could provide a mesoscale map of white matter axons across the entire macaque brain. A single brain could be imaged in ~4 days (**Table S3**). Existing efforts to map pathways in the primate brain at this resolution requires slicing the brain in 300 μm sections and laboriously assembling the resulting images into a coherent 3D volume (Xu et al. 2021 [DOI](#)).

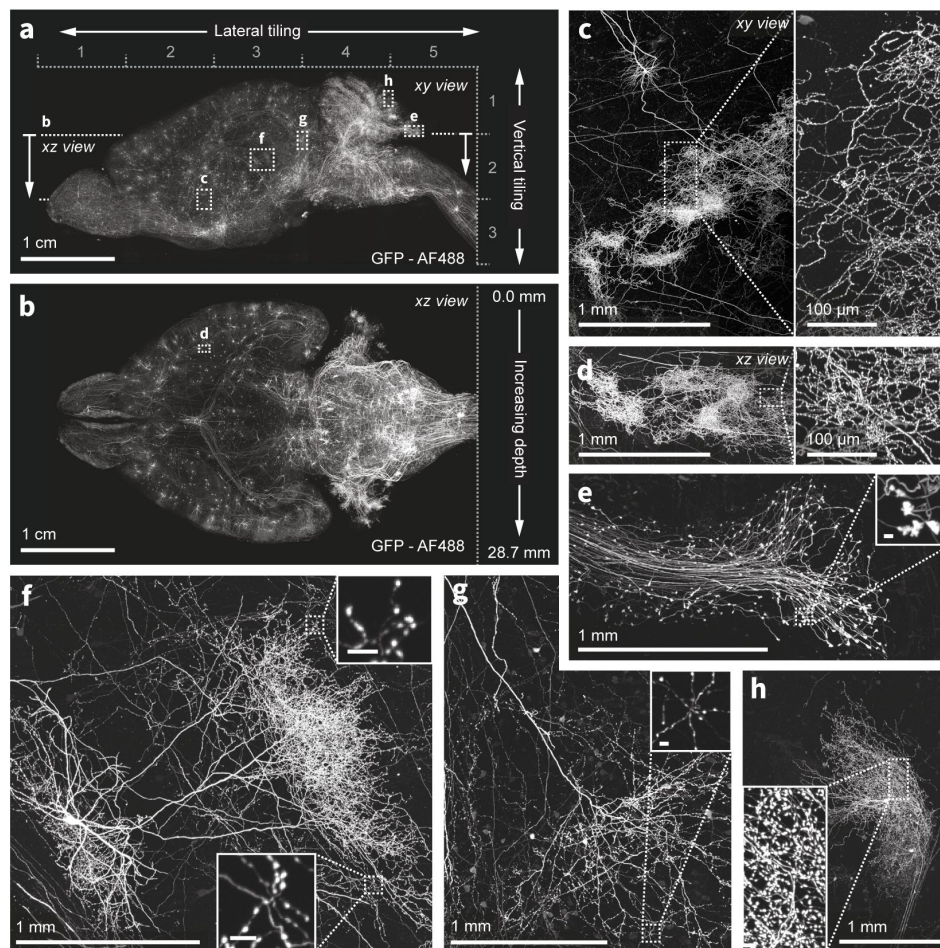


Figure 3

Nanoscale imaging of entire intact mouse brains.

(a-b) PHP.eB_pseudotyped pAAV-AiE2255m-minBG-iCre(R297T)-BGHpA, - an enhancer AAV that targets the 149 PVT-PT Ntrk1Glut subclass in the whole mouse brain taxonomy (Zizhen Yao et al. 2023 [dx.doi.org/10.17504/protocols.io.n92dpwjl5b/v1](https://doi.org/10.17504/protocols.io.n92dpwjl5b/v1)) driving the mutated iCre(R297T) recombinase, delivered retro-orbitally with a GFP reporter virus, sparsely labels cells brain wide, including neurons in the olfactory bulb, striatum, thalamus, midbrain pons and medulla. Intact mouse brains were expanded (3×) and imaged using the ExA-SpIM microscope ((Ouellette et al. 2023 [dx.doi.org/10.17504/protocols.io.n92dpwjl5b/v1](https://doi.org/10.17504/protocols.io.n92dpwjl5b/v1))). Due to a lack of sectioning, stable specimen mounting, fast imaging times, and minimal imaging distortions, adjacent imaging tiles are easily aligned in overlapping regions (**Supplementary Videos 1-2**). **(c-h)** ExA-SpIM imaging resolves dense axonal arbors and varicosities across various brain regions with high resolution and isotropy (**Supplementary Videos 3-6**). xy and xz views of a dense arbor in the striatum (c-d); mossy fiber axon terminals and boutons (**e**); an individual hippocampal CA3 neuron and its extensive local axons (**f**); long-range axonal projections of a medial septal neuron in the hippocampal formation (**g**); an individual basket cell in the cerebellum. Images are displayed as maximum intensity projections and inset scale bars correspond to 10 μm unless otherwise specified.

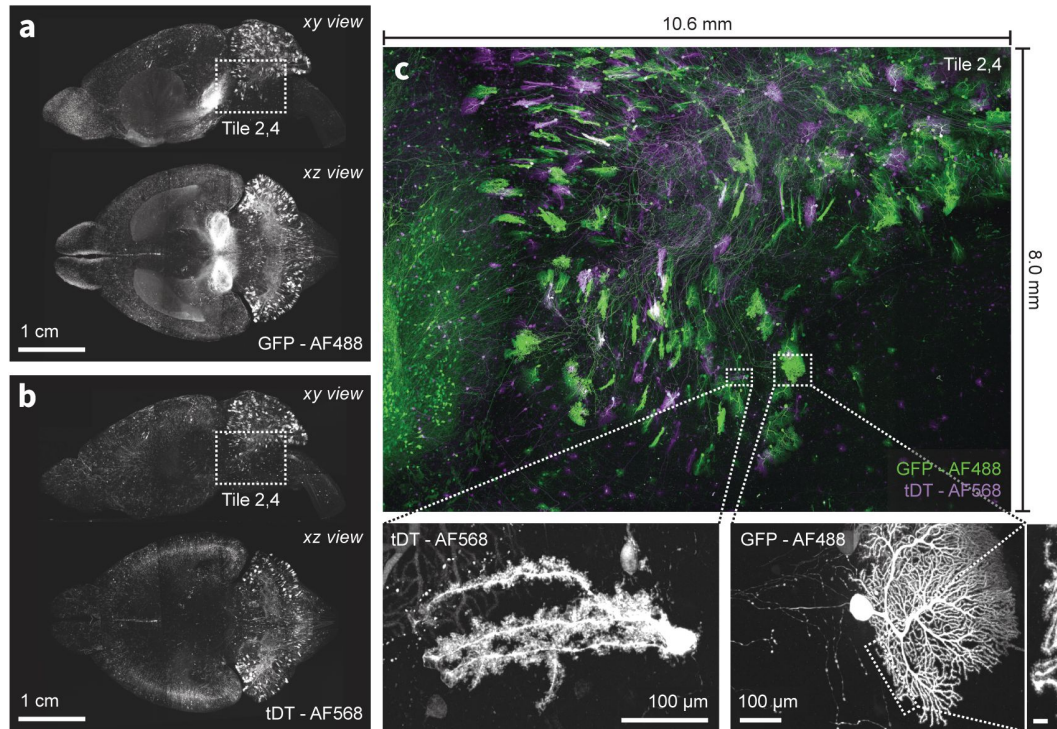


Figure 4

Multi-color imaging of centimeter-scale tissues with nanoscale resolution.

(a-b) An intact mouse brains was expanded (3×) and sparsely labeled neurons expressing GFP and tdTomato were imaged using the ExA-SPIM microscope ((Ouellette et al. 2023 <https://doi.org/10.17504/protocols.io.n92ldpwjx15b/v1>)). An overlay of both channels for a single tile is shown in **(c)**. The achromatic performance of the optics enabled diffraction-limited imaging in both color channels, as illustrated by the ability to resolve individual spines along the dendrites of Purkinje cells in the cerebellum. Images are displayed as maximum intensity projections and inset scale bars correspond to 10 μm unless otherwise specified.

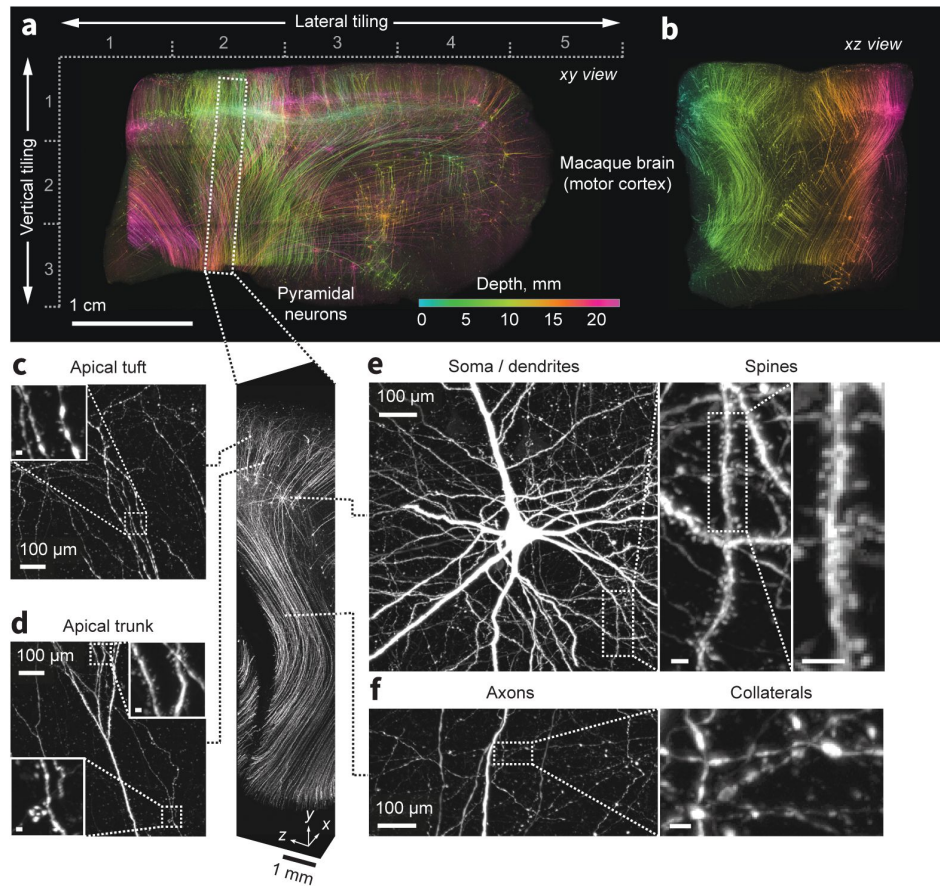


Figure 5

Expansion and imaging of a large volume of macaque brain.

A 1×1×1.5 cm block of macaque primary motor cortex was expanded (3×) and imaged on the ExA-SPIM (**Supplementary Videos 7-9**). Corticospinal neurons were transduced by injecting tdTomato-expressing retro-AAV into the spinal cord. **(a-b)** Maximum intensity projections of the imaged volume pseudo colored by depth. The axes descriptors in **(a)** indicate the 5×3 tiling used to image this volume. **(c-f)** Fine axonal and dendritic structures including descending axons, collaterals and dendritic spines are clearly discernable in the images throughout the entire volume. See **Supplementary Video 10**. Images are displayed as maximum intensity projections with the following thicknesses: **(a)** 23 mm, **(b)** 45 mm, **(c-f)** 1 mm.

Visualizing axons in human neocortex and white matter

Heavy chain neurofilaments comprise the internal scaffolds of long-range projection axons. Visualizing these neurofilaments with immunofluorescence could provide detailed information about axonal trajectories without the need for gene transfer methods, and which cannot be achieved with other methods, such as diffusion magnetic resonance imaging (Huang et al. 2021 [↗](#); Van Essen et al. 2019 [↗](#)) or high-resolution optical coherence tomography (Tianqi Li et al. 2019 [↗](#)). We next evaluated ExA-SPIM for imaging immunolabeled heavy chain neurofilaments in the human neocortex (**Figure 6** [↗](#)).

We cleared and expanded (4×) a ~1×1×0.01 cm piece of human neocortex (**Figure 6a** [↗](#)) and labeled the tissue with fluorescent SMI-32, which preferentially stains heavy chain neurofilaments. Individual axons and their trajectories are clearly visible through the 350 μm thick sample (**Figure 6b-d** [↗](#)). The larger axons (>1 μm) are well separated from each other in both the gray and white matter. In the white matter, axons are arranged as multiple intercalated populations coursing in different directions, rather than as homogeneous fascicles. These results demonstrate the feasibility of using ExA-SPIM for visualizing white matter tract axons and lay the foundation for scaling this data acquisition to multiple, thick tissue sections, and ultimately the entire human brain.

Discussion

Recent breakthroughs in histological methods (Richardson & Lichtman 2015 [↗](#); Spalteholz 1914 [↗](#); Dodt et al. 2007 [↗](#); Tsai et al. 2009 [↗](#); Hama et al. 2011 [↗](#); Becker et al. 2012 [↗](#); Ertürk et al. 2012 [↗](#); Chung & Deisseroth 2013 [↗](#); Ke et al. 2013 [↗](#); Susaki et al. 2014 [↗](#); Tainaka et al. 2014 [↗](#); Yang et al. 2014 [↗](#); Renier et al. 2014 [↗](#); Hou et al. 2015 [↗](#); Costantini et al. 2015 [↗](#); AU - Chi et al. 2018 [↗](#)), fluorescent labeling strategies (Kim et al. 2015 [↗](#); Susaki et al. 2020 [↗](#); Mao et al. 2020 [↗](#)), fast and sensitive cameras, and new microscopes (Huisken et al. 2004 [↗](#); Dodt et al. 2007 [↗](#); Wu et al. 2013 [↗](#); Kumar et al. 2014 [↗](#); Tomer et al. 2014 [↗](#); Economo et al. 2016 [↗](#); Narasimhan et al. 2017 [↗](#); Power & Huisken 2017 [↗](#); Migliori et al. 2018 [↗](#); Chakraborty et al. 2019 [↗](#); Glaser et al. 2019 [↗](#); Voigt et al. 2019 [↗](#); Voleti et al. 2019 [↗](#); Y. Chen et al. 2020 [↗](#); D. Chen et al. 2020 [↗](#); Wang et al. 2021 [↗](#); Xu et al. 2021 [↗](#); Zhang et al. 2021 [↗](#); Glaser et al. 2022 [↗](#); Xiaoli Qi et al. 2023 [↗](#); Vladimirov et al. 2024 [↗](#)) are revolutionizing our ability to study the molecular organization of cells and tissues with fluorescence microscopy.

For many applications it is necessary to reconstruct tissues with high resolution and contrast over large spatial scales, including cells, organs, or even entire organisms. For example, such multi-scale imaging is necessary to reconstruct individual neurons in the mouse brain, which can span many millimeters, but with axons that are often less than 100 nm thick. ExA-SPIM addresses this need for imaging of large tissue volumes with high resolution and contrast. The resulting image volumes are sufficient for manual neuron tracing, and preliminary data suggests that automated segmentation of thin axons is improved compared to previous approaches (Winnubst et al. 2019 [↗](#)). ExA-SPIM relies on tissue expansion and lower NA optics compared to microscopes that are typically used for high-resolution imaging. This approach has several advantages over imaging of cleared (non-expanded) tissues. First, optical engineering requirements are more forgiving for lenses with lower NA. As shown in (**Figure 1b** [↗](#)), lower NA lenses can have much higher etendues, fields of view, and much longer working distances. In other words, a large volumetric coverage (accessible resolvable voxels) is currently more feasible with lower NA lenses. In addition, these lenses are more easily corrected for common distortions, including field curvature, pincushion distortion, vignetting, and uniform resolution across the entire field of view (Yueqian & Herbert 2019 [↗](#)). These lenses can be paired with large format CMOS sensors that offer voxel rates

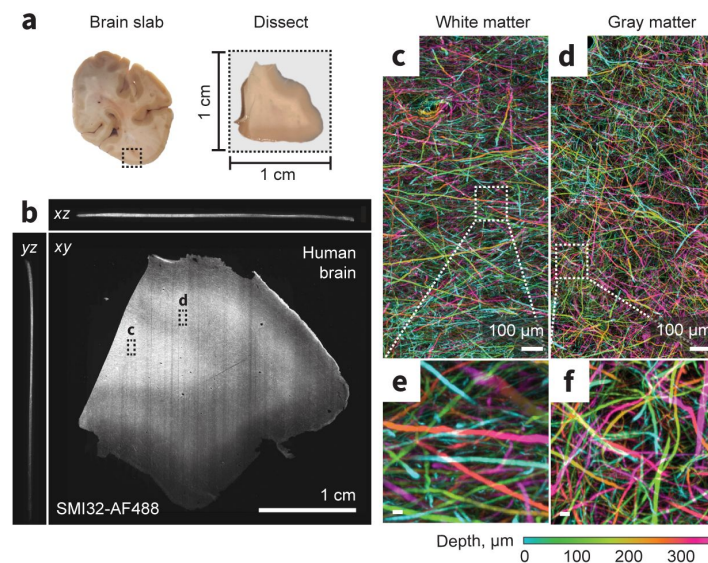


Figure 6

ExA-SPIM imaging of human tissue.

(a-b) A region from the medial temporal-occipital cortex was manually dissected into a $\sim 1 \times 1$ cm block, which was subsequently sectioned into ~ 100 μ m sections for tissue expansion (4 $\times$), labeling, and ExA-SPIM imaging. **(c-d)** Maximum intensity projection of a region of interest from white and gray matter, pseudo colored by depth. **(e-f)** Individual axons and their trajectories are clearly resolved with high contrast. Images are displayed as maximum intensity projections across 350 μ m **(b-f)**.

surpassing more commonly used scientific CMOS sensors. These technologies will continue to progress rapidly. For example, the recently announced 247 megapixel Sony IMX811 sensor (<https://www.sony-semicon.com/en/products/is/industry/rolling-shutter.html>) will be capable of delivering 12-bit images at up to 3 gigavoxels/sec.

Second, lower NA imaging is less sensitive to tissue-induced aberrations. Although clearing and expansion techniques render tissues transparent, remnant refractive index inhomogeneities still deteriorate image quality at larger imaging depths (Weiss et al. 2021). This is more noticeable at higher NA, because the rays entering the objective at higher angles have longer path lengths through the tissue and are more susceptible to aberrations. For lower NA systems the differences in path lengths between the extreme and axial rays are smaller. For example, spherical aberration (which may be a dominant aberration mode for imaging cleared and expanded tissues) increases with NA to the third power.

Third, the index of refraction gradients in tissue decrease with the third power of the expansion factor (proportional to density). As a consequence, tissue-induced aberrations reduce with the expansion factor to the third power. In contrast, the imaging path length through the tissue only increases with the expansion factor to the first power. These factors together explain the advantage of expansion-assisted imaging (Figure S4). Despite reduced sensitivity to aberrations and scattering, ExA-SPIM still exhibits a decrease in image quality at larger imaging depths (Figure S5). This can be reduced with dual-sided excitation (Dodt et al. 2007) as well as multi-view detection (Tomer et al. 2012).

One limitation of the current implementation of the ExA-SPIM is related to inefficient collection of signal photons. Given that signal collection scales approximately with NA^2 , signal collection is $10\times$ lower compared to a $NA = 1.0$ system ($(1/0.305)^2$). However, we find that the ExA-SPIM system can detect and resolve dim, nanoscale biological features with high signal-to-noise ratio (e.g., thin axons). Because of the large field of view, high SNR imaging requires $1000+$ mW lasers. However, light intensities are similar to traditional SPIM systems because the power is distributed over a ~ 1 cm wide light sheet. Under these imaging conditions photobleaching is not detrimental (i.e., $\sim 50\%$ reduction in intensity after ~ 800 repeated exposures), suggesting that even higher laser intensities could be used for higher SNR imaging (Figure S6). To improve signal collection and enable even higher sensitivity ExA-SPIM imaging, a custom lens with $NA = 0.5$, $FOV = 6.8$, and a 35 mm working distance in liquid has been designed and is being fabricated. This new lens will enable $\sim 50 - 150$ nm effective lateral resolution in $3-12\times$ expanded tissues, with $2.7\times$ improved light collection efficiency.

Fundamentally, contrast, effective resolution and imaging speed in expansion microscopy are limited by the density of fluorescent molecules labeling the structure of interest and the photon budget. ExA-SPIM allows distributing the technical burden between optics and tissue expansion (Figure S7). In practice, a target resolution should first be specified, corresponding to a viable combination of NA and tissue expansion, taking into consideration light collection efficiency (NA dependent) and required working distance (expansion factor dependent).

Although we have focused on the combination of new microscopy with tissue expansion, the ExA-SPIM microscope design will be useful for many additional imaging applications. For example, the field of view of the ExA-SPIM is sufficient to image a cleared mouse brain without the need for tiling. Cleared mouse brain datasets with $\sim 1 \mu m$ lateral resolution could be acquired at a speed of ~ 36 min/channel (Figure S8). In addition, the entire system could be converted to an inverted (Wu et al. 2013; Kumar et al. 2014) or open-top (McGorty et al. 2015; Glaser et al. 2017; Glaser et al. 2019; Barner et al. 2020; Glaser et al. 2022) architecture, which would enable large-scale imaging of tissue slabs with large aspect ratios that are up to 1 cm thick (Figure S9). Similarly, an ExA-SPIM system with a lower resolution metrology lens could easily be built (see

Table S3). These more mesoscale systems would provide extremely high imaging throughput (cm³ per day) and could pave the way for new large-scale neuroanatomy investigations of human and non-human primate tissues.

Finally, new types of applications may also require multi-scale microscopy. For example, expansion microscopy could provide a super-resolution view of interactions between immune cells and solid tumors. This application may require 10× expansion, for better than 100 nm effective resolution, to distinguish immunofluorescence in immune cells and tumor cells, throughout millimeter-scale tissue samples (10 mm after expansion). Given that 10× expanded samples are mechanically fragile and difficult to handle and section, the large field of view and reduced need for physical sectioning of ExA-SPIM would be highly advantageous for these demanding imaging experiments.

In summary, our new ExA-SPIM approach represents a new substrate for innovation within the realm of large-scale fluorescence imaging. By leveraging technologies from the electronics metrology industry, in combination with whole-mount tissue expansion, the system provides: (1) nanoscale lateral resolution, (2) over large centimeter-scale volumes, (3) with minimal tiling and no sectioning, (4) high isotropy, and (5) fast imaging speed.

Materials and methods

Microscope design

A parts list and CAD model are available at <https://github.com/AllenNeuralDynamics/EXA-SPIM-hardware>. A ZEMAX model is available at <https://github.com/AllenNeuralDynamics/EXA-SPIM-optics>. The system has two main optical paths (**Figure S10**). The first excitation path shapes and delivers the light sheet to the specimen. Laser light is provided by a custom-made laser combiner containing a 200 mW 405 nm laser (LBX-405-180-CSB-OE, Oxixus), 1000 mW 488 nm laser (Genesis MX488-1000 STM OPS Laser-Diode System, Coherent Inc.), 1000 mW 561 nm laser (Genesis MX561-1000 STM OPS Laser-Diode System, Coherent Inc.), and 1000 mW 639 nm laser (Genesis MX639-1000 STM OPS Laser-Diode System, Coherent Inc.) (**Figure S11**). Due to the large light-sheet, the ExA-SPIM system benefits from high excitation lasers powers (1000+ mW), resulting in light intensities typically used in SPIM microscopy. Photobleaching is not detrimental at these imaging conditions. An acousto-optic tunable filter (AOTFNC-400.650-TN, AA Opto Electronic) controlled by a RF driver (MPDS8C-D65-22-74.158-RS, AA Opto Electronic) is used to modulate each laser and control the output power of the combiner. The output beam from the combiner is ~1.2 mm in diameter, which is expanded to 13.3 mm using two relay lenses with $f = 3.6$ mm (LMPLANFLN 50×, Olympus) and $f = 40$ mm (HLB M PLAN APO 5×, Shibuya Optical). The light path is then reflected vertically by a third kinematic mirror and focused along one axis by an achromatic cylindrical lens (ACY254-050-A-ML, Thorlabs). The cylindrical lens is mounted in a motorized rotation mount (C60-3060-CMR-MO, Applied Scientific Instrumentation). This enables precise electronic control over the rotation of the light sheet within the specimen. Control over this parameter is important, as the field of view (~10.6 mm) and depth of field of the detection lens (<10 μm) require the sheet to be rotated with <1 deg precision. The light from the cylindrical lens is focused onto an electrically tunable lens (EL-16-40-TCVIS-20D-C, Optotune AG). The actuating surface of the electrically tunable lens is conjugated to the back focal plane of the excitation objective (1-290419, Navitar) through two large 20 mm galvanometric scanning mirrors (QS20X-AG, Thorlabs), three additional kinematic mirrors, and a final relay consisting of a first lens with $f = 200$ mm (AC508-200-A-ML) and second lens with $f = 300$ mm (AC508-300-A-ML). The two galvanometric mirrors are not conjugated to the back focal plane of the excitation objective and are used in tandem to tilt and translate the light sheet to be co-planar with the detection lenses' focal plane. Similar to the cylindrical lens rotation, the light sheet must also be tilted to <1 deg precision across the ~8.0 mm vertical height of the imaging field of view.

The excitation objective has a focal length of $f = 110$ mm, $NA = 0.133$, and back aperture diameter of ~ 29.26 mm, with which the preceding excitation optics, results in a Gaussian excitation light sheet with $NA \sim 0.1$. The intensity profile across the width of the light sheet is also Gaussian, with a full-width half-maximum (FWHM) of ~ 12.5 mm. This corresponds to a reduced (60%) light intensity at the edges of the field of view. The lens is mounted in a custom dipping cap which extends the effective working distance from 39 mm in air to ~ 52 mm in water. The 1-mm thick window on the dipping cap was sealed with UV curing optical adhesive (NOA86, Norland Products). The lens is oriented vertically, such that the light sheet is delivered downward into the immersion chamber which is filled with an immersion medium.

Fluorescence is collected with a high etendue electronics metrology lens (VEO_JM DIAMOND 5.0×/F1.3, Viewworks Co., LTD; jointly developed and fabricated by Schneider-Kreuznach). Whereas the spherical aberration introduced in the excitation path is negligible, the same dipping cap design would result in severe spherical aberration on the detection path, where the aperture was larger ($NA = 0.305$). Imaging at these higher apertures necessitated imaging lenses designed for direct immersion into a liquid mounting medium. The VEO_JM DIAMOND detection lens is designed for co-axial illumination, involving a 35 mm thick BK7 glass ($n = 1.52$) beam splitter mounted on the front object side of the lens. The beam splitter couples white light into the lens for bright illumination of electronic components during inspection. This was leveraged for aberration-free imaging into a liquid mounting medium.

We removed the beam splitter and replaced it with an optically equivalent thickness of the aqueous mounting medium (30 - 35 mm for refractive indices between 1.33 - 1.56) and glass (4 mm total, including the chamber window and fluorescence filter). This scheme avoids the spherical aberrations that would otherwise plague imaging with an air-immersion lens into a non-air-based medium. The lens is positioned horizontally on the optical table outside of the immersion chamber. The lens images the specimen through a $\lambda/10$ 50 mm diameter VIS-EXT coated fused silica window that is 3 mm thick (#13-344, Edmund Optics). The window was glued to the immersion chamber using UV curing optical adhesive (NOA86, Norland Products). A custom 44 mm diameter 1 mm thick multi-bandpass fluorescence filter (ZET405/488/561/640nm2, Chroma) is inserted on the outer side of the immersion chamber and held in place with a retaining ring (SM45RR, Thorlabs). Placing the filter on the object side of the lens results in a cone angle of incidence of 17.8 deg. This results in a spectral shift and broadening of transmitted light, which was modeled to account for adequate suppression of scattered light at the excitation laser wavelengths.

The lens is attached to the camera (VP-151MX-M6H00, Viewworks Co., LTD) using a custom mechanical assembly that enables tipping and tilting of the lens and camera relative to the immersion chamber window. The camera uses the Sony IMX411 sensor, which features a 14192×10640 array of 3.76 μm pixels. The 5.0× magnification of the detection lens implies sampling at 0.75 μm in the specimen plane. Because the detection lens is not infinity-corrected, it is not possible to change tube lenses to change the pixel sampling. Therefore although the optical resolution of the detection lens is ~ 1.0 μm , the sampling limited resolution based on the Nyquist criterion is 1.5 μm . However, it is worth noting that Viewworks Co. produces cameras with pixel-shifting technology (See the pixel-shifting section at: <https://vision.viewworks.com/en/technology>), which would enable finer sampling if needed. The cooling fan on the camera was replaced with a quieter fan operating at 2400 rpm (Noctua NF-A6x25 FLX, Premium Quiet Fan, 3-Pin, 60 mm, Noctua). Using an alignment laser, the entire assembly was aligned to the window. The entire assembly is also mounted on a rail system (XT95, Thorlabs) that enables precise axial alignment of the assembly relative to the chamber window.

The sensor can be operated with 12, 14, or 16-bit analog to digital (A/D) conversion, each of which provides a trade-off between noise and data throughput. When operated with 14-bit A/D, the line time of the sensor is 20.15 μs . With 14192 pixels per row on the sensor, this corresponds to an

imaging speed of 703×10^6 voxels/sec.

The line times and imaging speeds at 12 and 16-bit are 15.00 and 45.44 μ s respectively, corresponding to 946 and 312×10^6 voxels/sec. This contrasts with a state-of-the-art sCMOS sensor with 2048 pixels per row, where even at the fastest 4.89 μ s line time, the imaging speed is only 418×10^6 voxels/sec. In other words, the Sony IMX411 sensor provides more pixel parallelization within each row. This enabled twice the voxel rate with four times the pixel dwell time. See **Note S1** for further discussion on comparing the speed and sensitivity of cameras.

The specimens, in this case an expanded gels, were mounted in a customized holder, which was attached to a motorized XY stage (MS-8000, Applied Scientific Instrumentation) and Z stage (LS-100, Applied Scientific Instrumentation), for scanning and tiling based image acquisition.

Microscope control

The microscope was controlled using a high-end desktop workstation (SX8000, Colfax Intl). The workstation was equipped with a motherboard with five PCIe 4.0x16 and one PCIe 4.0x8 slots (X12DAI-N6, Supermicro), most of which are required for the various electronics needed to control the ExA-SPIM. One slot was used for the frame grabber (Coaxlink Octo, Euresys), which streams imaging data from the camera onto a fast local 12.8 TB NVME drive (7450 Max 12800GB 3 DWPD Gen4 15mm U.3, Micron). A second slot was used for the data acquisition (DAQ) card (PCIe-6738, National Instruments) used for generating the various digital and analog voltage signals. A third slot was used for the high-speed network interface card (ConnectX-5 EN MCX515A-CCAT QSFP28 Single Port 100GbE, Mellanox) to transfer data off the local NVME drive, over the network, and onto a networked server. A fourth slot was occupied by the workstation GPU (A4000, NVIDIA). The workstation was also connected to a controller (TG-16, Applied Scientific Instrumentation) via a USB connection. The TG-16 controller was equipped with cards for controlling the motorized X, Y, and Z stages, the electrically tunable lens, and the cylindrical lens rotation mount. Finally, each laser within the combiner was connected to the computer via a USB connection, as well as the RF driver of the AOTF. The DAQ acted as the master controller of the entire system. Analog output voltages were used to drive or trigger the electrically tunable lens, two scanning galvanometric mirrors, camera, scanning stage, and the RF outputs to the AOTF for each laser.

Acquisition software

The microscope was controlled using custom software written in Python <https://github.com/AllenNeuralDynamics/voxel> and was operated on the Windows 10 operating system. The software could control and configure all of the electronically controlled hardware devices and uses Napari (<https://napari.org>) as the graphical user interface (GUI) for image streaming and visualization. An imaging experiment consisted of a series of nested loops, which included looping over the total number of frames within a given tile, looping over the total number of channels within a tile, and finally tiling in two dimensions to cover the entire tissue volume. A multiprocessing, double buffering scheme was used to capture a tile while the previous tile was being transferred over the network to longer term storage. All hardware from Applied Scientific Instrumentation was controlled using a custom API written in Python <https://github.com/AllenNeuralDynamics/TigerASI>. The AOTF was controlled using a custom API written in Python <https://github.com/AllenNeuralDynamics/aaopto-aotf>.

To achieve robust and reliable acquisition at the speed required by the ExA-SPIM, we used two data streaming strategies. The first used the open-source eGrabber frame grabber Python API (<https://www.euresys.com>) and open-source ImarisWriter library (Beati et al. 2020). This option enabled high-speed data streaming with online lossless compression (LZ4 with bit shuffling) and real-time writing of a multi-resolution pyramid, both of which help streamline downstream data storage and handling. The second strategy used Acquire (<https://github.com/acquire-project>)

(Clack et al. 2023 [↗](#)). Acquire is a new state-of-the-art microscope acquisition project which enables the ExA-SPIM system to stream data at the required rate, directly to OME-Zarr, with either ZStandard or LZ4 compression, and a variable chunk size.

On average, with the 14-bit A/D setting of the camera (MSB packed into 16 bits), the lossless compression ratio using either ImarisWriter or Acquire was $\sim 2\text{--}4\times$. This reduced the overall effective data rate of the system and eased network transfers of the data to centralized storage. A schematic of the acquisition pipeline is shown in **Figure S12** [↗](#).

Point spread function quantification

To quantify the point spread function of the ExA-SPIM microscope we imaged fluorescent $0.2\text{ }\mu\text{m}$ TetraSpeckTM microspheres (Invitrogen, cat no. T7280, lot no. 2427083) using 561 nm excitation. A cube of expanding hydrogel containing a 10% (v/v) microbead solution was prepared 40 μL of microspheres solution was added to 360 μL of activated Stock X monomer solution, as described in (Asano et al. 2018 [↗](#)). The solution was briefly vortexed, and carefully pipetted into an array of 2 mm^3 wells in a silicone mold. The mold was placed in a sealed petri dish containing damp Kimwipes® to maintain humidity and incubated at 37°C for 2 hours. After incubation, the polymerized bead phantoms were placed into $0.05\times$ SSC buffer for at least 24 hrs to expand and equilibrate prior to imaging. After volumetric imaging, the resulting imaging stack was analyzed using custom written Python code <https://github.com/AllenNeuralDynamics/exa-spim-characterization> [↗](#). The results shown in **Figure 2** [↗](#) are averaged from $>10,000$ individual beads.

Field curvature quantification

We used the field curvature quantification methods developed for the Benchtop mesoSPIM (Vladimirov et al. 2024 [↗](#)). Briefly, the field curvature was measured using a high-precision Ronchi ruling with 120 lines per mm (62-201, Edmund Optics). The ruling was mounted into the system, and aligned to be normal (i.e., flat) with respect to the imaging path. The ruling was trans-illuminated using light-emitting diodes (LEDs) at 450 nm (M450LP2, Thorlabs), 530 nm (M530L4, Thorlabs), 595 nm (M595L4, Thorlabs), and 660 nm (M660L4, Thorlabs). To provide uniform illumination, the LEDs were collimated to a diameter of $\sim 20\text{ mm}$ and passed through a diffuser (ED1-C20, Thorlabs). Image stacks were captured by scanning the Ronchi ruling in $1\text{ }\mu\text{m}$ steps through the imaging path focal plane (2 mm total scan range) with illumination at either 450 nm, 530 nm, 595 nm, or 660 nm. The resulting image stack was split into a 16×16 grid of regions of interest (ROI), each $887\times 665\times 2000$ pixels. Within each ROI, the contrast of each frame in the stack was calculated using the 5th and 95th percentiles of intensity within the frame, where the contrast, $C = (I_{\text{max}} - I_{\text{min}})/(I_{\text{max}} + I_{\text{min}})$. The resulting contrast versus depth curve was fit to a normal distribution to extract the depth (i.e., index) corresponding to maximum contrast. The indices were radially averaged across the 16×16 grid of ROIs, yielding an estimate of the imaging lens field curvature within each of the four tested wavebands. Custom Python scripts were used to run the analysis <https://github.com/AllenNeuralDynamics/exa-spim-characterization> [↗](#).

Lens distortion quantification

The lens distortion was measured using a target (62-950, Edmund Optics) with $125\text{ }\mu\text{m}$ diameter dots spaced every $250\text{ }\mu\text{m}$ in a grid pattern. The target was mounted and trans-illuminated using LEDs in the same manner as the field curvature quantification. A single image was acquired with the target at the focal plane of the imaging lens. The resulting image was quantified by first segmenting and calculating the centroid of each dot. The calculated position of each dot was then compared to the theoretical dot position. The percent distortion for each dot was defined as the difference between the experimental and theoretical positions, normalized by the theoretical position. The resulting distortion values were radially averaged, yielding the lens distortion as a function of position from the center of the lens field of view. Custom Python scripts were used to run the analysis <https://github.com/AllenNeuralDynamics/exa-spim-characterization> [↗](#).

Camera sensitivity comparison

A bead phantom (see point spread function measurement) was used to compare the sensitivity of the large-format CMOS camera and a scientific CMOS camera (Orca Flash V3, Hamamatsu). A single bead was first imaged with the scientific CMOS camera, varying the effective pixel rate or imaging speed. The scientific CMOS camera was then removed from the microscope and replaced with the large-format CMOS camera.

The same bead was located and imaged again at various effective pixel rates. The bead was located in all image stacks, and the signal-to-noise ratio of the bead was quantified as the signal of the bead, minus the average background signal, divided by the background noise. A custom written Python code was used to run the analysis <https://github.com/AllenNeuralDynamics/exa-spim-characterization>. It is worth noting that the Orca Flash V3 is an older scientific CMOS camera. Newer cameras, such as the Orca BT Fusion with improved noise characteristics (1.0 e- versus 1.6 e- read noise and 95% versus 85% peak quantum efficiency) are now available. However, these differences would not significantly change the results shown in **Figure 2e**. **Note S1** contains a deeper discussion on comparing the speed and sensitivity of cameras.

Local data storage and handling

Data was streamed using the ImarisWriter or Acquire onto a local NVME drive in the acquisition workstation. Upon the completion of a tile, the resulting files were transferred over the high-speed network using xcopy (<https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/xcopy>) onto a 670 TB enterprise storage server from VAST data. This occurred in parallel with the acquisition of the next imaging tile. Upon completion of the transfer, the tile was deleted off the acquisition workstation's NVME drive.

Image compression and cloud transfer

The raw data for each color channel consisted of a set of overlapping 3D image tiles. For the ImarisWriter acquisition workflow, the images were stored in Imaris IMS file format (<https://imaris.oxinst.com/support/imaris-file-format>), an HDF5-based format which enables fast parallel writing to local storage. However, this format is not well suited to being archived in cloud (object) storage since accessing an arbitrary data chunk requires seeking a file. Thus, the data was converted to OME-Zarr format, which supports parallel read-write over the network and has a rich metadata structure (Moore et al. 2021; Moore et al. 2023). OME-Zarr is also integrated with visualization and annotation tools used for downstream visualization and analysis, including Horta Cloud (<https://github.com/JaneliaSciComp/hortacloud>) and Neuroglancer (<https://github.com/google/neuroglancer>). With up to four channels, the storage footprint for a single dataset can reach hundreds of terabytes. Therefore, the storage ratios and compression/decompression speeds of various lossless codecs were compared. Blosc ZStandard yielded the highest storage ratios, with compression speeds comparable to LZ4 at lower 'clever' settings (e.g., 1-3). Dask (<https://www.dask.org>) was used to parallelize the compression and OME-Zarr writing over a high-performance computing (HPC) cluster consisting of 16 nodes, each with 32 Intel CPUs with Advanced Vector Instructions 2 (AVX2) and 256 GB RAM. Image chunks were read in parallel from high-bandwidth network storage, compressed in memory, and written directly to AWS S3 and Google Cloud Storage buckets. Combined throughput (chunk read, compress, write) reached over 2 GB/s, with execution time dominated by read-write I/O operations. The image compression and cloud transfer code is available at <https://github.com/AllenNeuralDynamics/aind-data-transfer>. For the Acquire acquisition workflow, datasets were streamed directly to the OME-Zarr format.

Image stitching

ExA-SPIM image stitching used a combination of on-premises and cloud-based resources. Datasets were first converted from OME-Zarr to N5 and stitched using BigStitcher on the Google Cloud Platform (GCP) (Hörl et al. 2019 [\[1\]](#)). Tile placement transformations were based on interest points. The interest point based tile registration consisted of three steps: identification of interest points, finding corresponding interest points between tiles, and optimizing for tile transformation parameters with respect to the distance of corresponding interest points. We first performed a translation only stitching routine, followed by a full affine transformation optimization. The affine transformation was regularized by a rigid transformation and corner tiles were kept fixed to prevent the global scaling of the sample and divergent solutions at the corner tiles. As a rule of thumb, several thousand interest points per overlapping region were required for a reliable identification of correspondences and successful optimization. Once the alignment transformations were calculated, the tiles were fused using the on-premises HPC into a single contiguous volume in the N5 format. A representative dataset (~100 TB in raw uncompressed size) can be fused in ~24 hours using 16 nodes (32 cores each, 480 total), with 16 GB of RAM per core (7.68 TB total) and an output chunk size of 256, 256, 256 pixels. 2 cores per node are reserved as overhead per spark worker (i.e., 32 total for 16 nodes). Fused datasets along with the raw tiled datasets and the tile placement transformations are deposited in an Amazon Web Services S3 and open-data data bucket. The N5 datasets were converted to OME-Zarr for visualization with Neuroglancer, or KTX (<https://registry.khronos.org/KTX> [\[2\]](#)) for visualization with HortaCloud. The current pipeline involves several file format conversion steps and transfers between on-premises and cloud-based platforms. Future pipelines will standardize around OME-Zarr and be completely operable in the cloud. Only the initial data conversion and compression step would be computed on-premises.

Image visualization and annotation

To generate whole-brain single neuron reconstructions we use HortaCloud, an open-source streaming 3D annotation platform enabling fast visualization and collaborative proofreading of terabyte-scale image volumes. Human annotators proofread stitched image volumes using HortaCloud in a web browser on a personal workstation. Starting from the soma, the axonal and dendritic arbors were traced through all terminals by laying down connected points along the neurite, producing a piecewise-linear approximation of neuronal trees (Economo et al. 2016 [\[3\]](#)).

Whole mouse brain tissue processing

Briefly, the samples were cleared, immunolabeled and expanded as described below. Detailed protocols for preparing cleared and expanded brains are available at [(Ouellette et al. 2023 [\[4\]](#)), [dx.doi.org/10.17504/protocols.io.n92ldpwjxl5b/v1](https://doi.org/10.17504/protocols.io.n92ldpwjxl5b/v1) [\[5\]](#)].

Viral labeling in mice

Adult transgenic Cre driver mice between ages p21 to p35 received systemic injections, via the retro-orbital sinus, of a 100 μ L mixture of Cre-dependent Tet transactivator (AAV-PHP-eB_Syn-FlexTRE-2tTA, typical dose 6.0×10^8 gc/mL) and a reporter virus (AAV-PHP-eB_7x-TRE-tdTomato, typical dose 1.8×10^{11} gc/mL) (Addgene plasmid id: #191210 and, #191207). Viral vector-mediated recombination was achieved by retro-orbital injection of AiP1999 - pAAV-AiE2255m-minBG-iCre(R297T)-BGHpA (Addgene id: 223843) along with the two aforementioned vectors. The viral titers of the tTA virus used were empirically adjusted based on the Cre driver line to yield sparsely labeled brains. Viruses were obtained from either the Allen Institute for Brain Sciences viral vector core, the University of North Carolina, BICCN-Neurotools core and were prepared in an AAV buffer consisting of 1 $\times$ PBS, 5% sorbitol, and 350mM NaCl.

Collection

All experimental procedures related to the use of mice were approved by the Institutional Animal Care and Use Committee of the Allen Institute for Brain Science, in accordance with National Institutes of Health (NIH) guidelines. Four weeks after viral transfection, mice (~p70) were anesthetized with an overdose of isoflurane and then transcardially perfused with 10 mL 0.9% saline at a flow rate of 9 mL/min followed by 50 mL 4% paraformaldehyde in PBS at a flow rate of 9 mL/min. Brains were extracted and post-fixed in 4% paraformaldehyde at room temperature for 3-6 hours and then left at 4°C overnight (12-14 hours). The following day, brains were washed in 1× PBS to remove all traces of excess fixative.

Delipidation

Whole brain delipidation was performed in two stages. First, brains were dehydrated through a gradient of tetrahydrofuran (THF) in deionized water at 4°C and then delipidated in anhydrous dichloromethane (DCM) at 4°C. The brains were rehydrated into water through a gradient of THF and then placed in 1× PBS. Second, whole brains were transferred from 1× PBS to a biphasic buffer (SBiP) for 5 days at room temperature and then rinsed in a detergent buffer (B1n) for 2 days.

Immunolabeling

Delipidated brains were equilibrated in a detergent buffer (PTxw) and then incubated in PTxw containing the primary antibody (10 µg/brain) at room temperature for 11 days. After thorough washing in PTxw, a solution of the secondary antibody (20 µg/brain) was added for 11 days at room temperature. Brains were then rinsed thoroughly with PTxw and transferred to 1× PBS.

Gelation and expansion

Immunolabeled brains were equilibrated in MES buffered saline (MBS) followed by incubation in acryloyl-X SE (AcX) at 4°C on wet ice for 4 days. The AcX solution was then rinsed off with 1× PBS and the brain was then transferred to a solution of StockX activated with VA-044 at 4°C on wet ice for 4 days. After StockX incubation, whole brains were placed in a polymerization chamber and filled with activated StockX solution. The chamber was sealed with a coverslip, placed in an inert atmosphere of N₂, and baked at 37°C for 4+ hours until hydrogel formation. The hydrogel was then digested with proteinase K for 10+ days, until the tissue cleared. Upon completion of digestion, the brain was rinsed with 1× PBS and expanded in 0.05× saline sodium citrate (SSC) until 3× expansion was achieved.

Macaque brain tissue processing

The process for preparing the macaque brain samples is described below.

Spinal injection procedure and motor cortex collection

To retrogradely label corticospinal neurons in the hand-wrist and trunk regions of primary motor cortex in a pigtail macaque, we injected a retro AAV vector (rAAV2-CAG-tdTomato; Addgene plasmid #59462 packaged in-house, titer of 1.88×10^{13}) into the left lateral funiculus and ventrolateral part of the gray matter in the C6/C7 spinal segments. All animal procedures have been approved by the University of Washington IACUC committee and conformed to the NIH's Guide for the Care and Use of Laboratory Animals.

An 11 year and 4 months old female *Macaca nemestrina* (10.55 kg) designated for the tissue distribution program was anaesthetized with isoflurane after an initial sedation with ketamine. The monkey was paralyzed with a neuromuscular blocker and artificially ventilated. The animal was monitored by a trained surgical technician for pulse oximetry, body temperature, ECG, blood

pressure, capnography, and inspired oxygen. External thermal support was provided for the duration of the surgery, and an intravenous line for i.v. drug and isotonic fluid administration, a urethral catheter was inserted to maintain fluid volume and physiological homeostasis. Under aseptic conditions, a partial laminectomy of the C5-C7 vertebrae was performed to expose the left dorsal surface of the cervical enlargement.

Using a stereotaxic manipulator (Kopf Instruments, Tujunga, CA) on a custom frame, targeted microinjections were performed with a Nanoject II (Drummond Scientific, Broomall, PA). A glass pipette with a broken tip (barrel of tip = 50 $\times$ m), filled with rAAV2-CAG-tdTomato, was inserted into the spinal cord through small longitudinal incision of the dura. Using the dorsal root entry points as a guide, 7 injection tracts spanning dorsal-ventral were used to target the lateral funiculus and ventrolateral part of the gray matter in the C6/C7 spinal segments. Each tract had five injections (138 nL of virus at 23 nL/second) positioned 100 μ m apart spanning -4.1 to -3.7 mm from the surface of the cord. The seven tract locations spanned 2.2 mm anterior-posterior and were in two rows (1 mm apart), evenly spread with slight adjustments to avoid hitting the vasculature. A one-minute wait period was used prior to each first injection within the tract, a two-minute wait period after each injection before moving the pipette and a five-minute wait period before removing the electrode after the final injection for each tract. The injector's efficacy at ejecting virus was confirmed between each injection tract.

After the injections were complete, artificial dura was placed over the durotomy, the musculature and skin were sutured, and an anti-paralytic agent (atropine) was delivered. Post-operative monitoring and care were performed to minimize pain and distress. Thirty days after the original injections, the animal was anaesthetized as described above, and then euthanized using a lethal dose of pentobarbital solution. After death, the animal was perfused transcardially with sodium-free oxygenated ice-cold artificial cerebrospinal fluid (NMDG-aCSF in mM): 92 NMDG, 25 glucose, 30 NaHCO₃, 20 HEPES, 10 MgSO₄, 2.5 KCl, 1.2 NaH₂PO₄, 0.5 CaCl₂, 3 sodium pyruvate, 2 thiourea, 5 sodium ascorbate. After perfusion, the brain was removed and the right hemisphere trunk and hand wrist subregions of primary motor cortex were dissected and stored in NMDG-saline on ice. A portion (2 cm³) of the motor cortex was further sub-dissected and stored in freshly made 4% paraformaldehyde in 0.1 M phosphate-buffered saline for later processing.

Delipidation, immunolabeling, gelation, and expansion

Macaque tissue was processed in the same manner as whole mouse brains.

Human brain tissue collection, delipidation, labeling, gelation, and expansion

Deidentified postmortem adult human brain tissue (61 year old male, Hispanic, no known history of neuropsychiatric or neurological conditions) was obtained with permission from next-of-kin by the San Diego Medical Examiner's Office. Tissue procurement was reviewed by the Western Institutional Review Board (WIRB) and did not constitute human subject research requiring Institutional Review Board (IRB) review, in accordance with federal regulation 45 CFR 46 and associated guidance. Postmortem tissue collection was performed in accordance with the Uniform Anatomical Gift Act described in Health and Safety Code §§ 7150, et seq., and other applicable state and federal laws and regulations.

Tissue was manually sliced into 1 cm coronal slabs, flash frozen with liquid nitrogen, vacuum sealed and stored at -80°C by the Allen Institute Tissue Processing Team. One slab from the occipital pole was drop-fixed in 4% PFA for approximately 12 hours at 4°C, and tissue regions containing visual cortex were dissected into ~1 cm blocks for histological processing. Individual blocks were then SHIELD-fixed (Lifecanvas Technologies) prior to sectioning at 100 μ m on a sliding freezing microtome to further protect protein antigenicity and tissue architecture.

Individual free-floating sections were then passively delipidated (LifeCanvas Technologies) for one week prior to immunolabeling. Delipidated sections were subsequently immunolabeled for SMI-32, which identifies heavy chain neurofilaments that make up axon scaffolds in long-range projection neurons. Free-floating sections were blocked in NGSTU (5% goat serum, 0.6% Triton X-100, 4M urea in 1× PBS) overnight, incubated with a primary antibody (rabbit anti-neurofilament 200, Sigma Aldrich N4142) diluted 1:500 in NGSTU + 0.02% sodium azide for 5 days, followed by a secondary antibody incubation (goat anti-rabbit AF488, ThermoFisher A-11034) diluted 1:100 in NGST (5% goat serum, 0.6% Triton X-100 in 1× PBS) for 4 days. Due to the thinness (100 μ m) of this tissue compared to an entire mouse brain, the gelling protocol used varied slightly from the whole brain gelling protocol described above. Most notably, the thermal initiator used was ammonium persulfate (APS) instead of VA-044, the tissue was polymerized at room temperature for 3 days, and the resulting tissue-hydrogel matrix was digested using a 1:50 concentration of proteinase-k in buffer (5% Sodium Dodecyl Sulfate, 5% Triton X-100, 10% 1M TRIS pH8).

Sample mounting

The expanded samples were trimmed to produce smooth edges and then were placed in a custom-built anodized imaging chamber. The sample chamber assembly was performed in a large bath of the expansion and imaging solution (0.05× SSC). The smooth edges of the hydrogel were placed against the chamber panels corresponding to the excitation and emission path and then the chamber removed from the bath. A warm (55°C) solution of 2% agarose, in the same 0.05× SSC as used during expansion, was carefully poured into the chamber space behind the hydrogel for structural rigidity during imaging and let cool to room temperature until solid. The chamber was then sealed and placed in 0.05× SSC for equilibration overnight before imaging. The detailed protocol for mounting the expanded hydrogels is available on protocols.io: [(Ouellette et al. 2023 <https://www.protocols.io/view/whole-mouse-brain-delipidation-immunolabeling-and-cmqbu5sn?step=12>)].

Data availability statement

Imaging datasets for this paper are available at:

s3://aind-open-data/exaSPIM_708373_2024-04-02_19-49-38

s3://aind-open-data/exaSPIM_671477_2024-01-08_17-03-04

s3://aind-open-data/exaSPIM_Z13288-QN22-26-036_2023-03-26_09-27-21

s3://aind-open-data/exaSPIM_H17.24.006-CX55-31B_2023-05-11_14-59-09

Instructions for how the data is organized and how it can be accessed is available at: <https://allenneuraldynamics.github.io/data.html>.

Code availability statement

Code is available as listed below. Please see the methods sections for additional usage details.

Microscope control: <https://github.com/AllenNeuralDynamics/voxel>

Acquire video streaming: <https://github.com/acquire-project>

Hardware files: <https://github.com/AllenNeuralDynamics/exa-spim-hardware>

Optical files: <https://github.com/AllenNeuralDynamics/exa-spim-optics> 

Characterization scripts: <https://github.com/AllenNeuralDynamics/exa-spim-characterization> 

Compression and cloud transfer: <https://github.com/AllenNeuralDynamics/aind-data-transfer> 

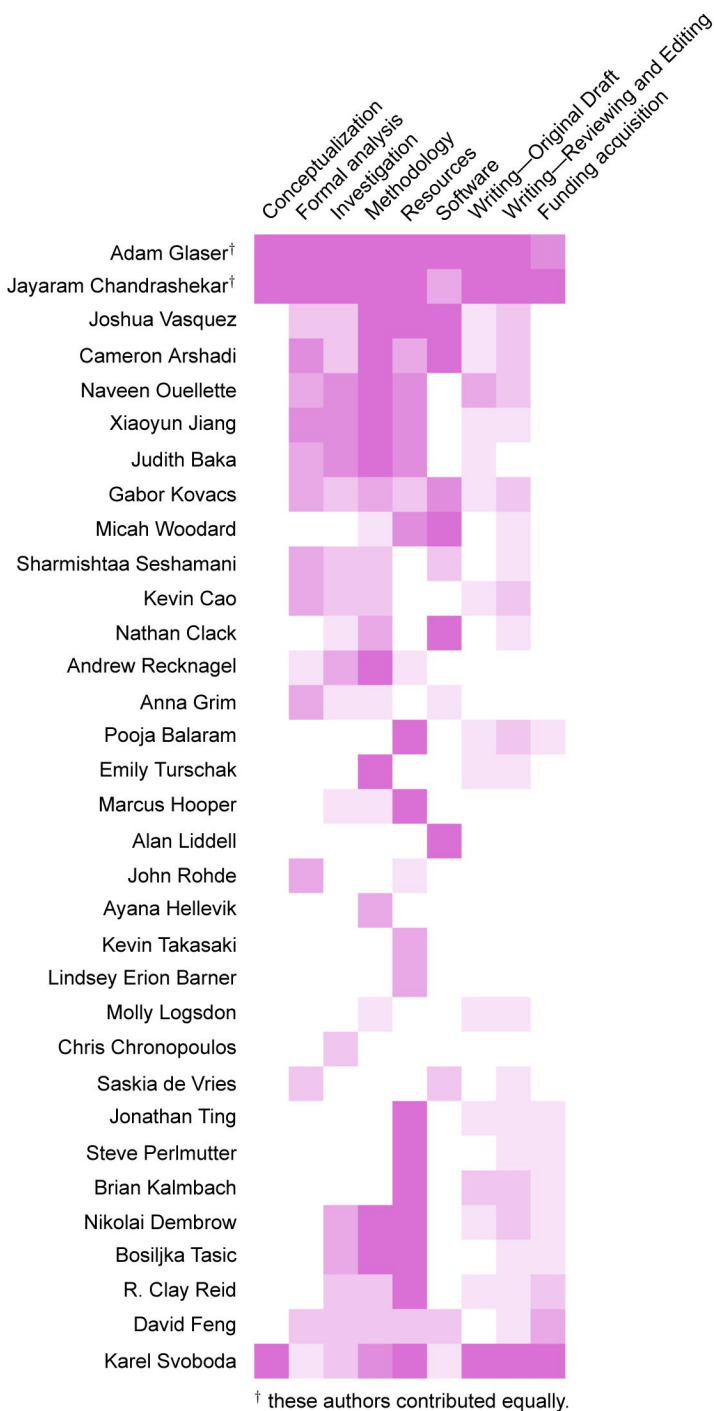
Acknowledgements

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Ethics declarations

A.G., J.C., and K.S. have filed for a patent on aspects of the ExA-SPIM system.

Author contributions



J.T. devised and validated the attenuated iCre mutant (R297T) for improved fidelity of recombination in the context of cell type specific enhancer AAV vectors.

Supplementary Material

Note S1. Comparisons of theoretical camera sensitivity and speed

Unlike existing SPIM systems, the ExA-SPIM system uses the large-format SONY IMX411 CMOS sensor with different specifications from sCMOS cameras. With respect to sensitivity, the read noise is the most important specification. While the read noise is ~ 1 e^- for a sCMOS camera, it is 6.89, 4.57, and 3.48 e^- for 12, 14, and 16-bit readout modes for the SONY IMX411 sensor. The signal-to-noise ratio (SNR) of a camera sensor can be calculated as:

$$SNR = \frac{QE \times S}{\sqrt{QE \times (S + B) + R^2}} \quad (2)$$

where, QE is the quantum efficiency of the sensor, S is the signal in photons, B is the background signal in photons, and R is the readout noise in e^- . Plots for a sCMOS and the Sony IMX411 sensor are shown in **Figure S12** [↗](#). These plots yield insights into for an equivalent number of photons, how the SNR of the sensors would differ, or how many photons each sensor would require to achieve the same SNR. In general, the sCMOS provides much greater SNR at low-light levels (i.e. < 50 photons), with diminishing benefits for higher photon levels, especially when compared to the 14, and 16-bit modes of the SONY IMX411 sensor. However, it is also important to consider the SNR as a function of the imaging speed (i.e., pixels/sec) of the sensor (**Figure 2e** [↗](#)). For example, when operated with 14-bit readout, the line time of the Sony IMX411 sensor is 20.15 μs . With 14192 pixels per row on the sensor, this corresponds to an imaging speed of 703×10^6 voxels/sec. This contrasts with a traditional sCMOS sensor that only contains 2048 pixels per row, where even at the fastest 4.89 μs line time, the imaging speed is only 418×10^6 voxels/sec. In other words, the Sony IMX411 sensor provides more pixel parallelization within each row. This enables twice the voxel rate with four times the pixel dwell time (i.e., four times the collected signal). It is important to note that sCMOS sensors provide a continuous adjustment of the sensor line time, to precisely trade-off between sensitivity and imaging speed, whereas the Sony IMX411 sensor only enables adjustment at three discrete setpoints, provided by changing the readout mode of the sensor between 12, 14, and 16-bit.

$$SNR = \sqrt{P}$$

where P (photons) was calculated by converting the arbitrary intensity units (a.u.) to photons using the gain factor, K (e^- /a.u.) and quantum efficiency, QE (e^- /photon) of the camera sensor.

Figure S1

Comparison between standard cleared-tissue and expansion-assisted SPIM.

(a-c) A traditional cleared-tissue SPIM system uses standard 100 mW or less lasers, a scientific CMOS camera, and life sciences objectives with higher NA and <10 mm working distance. In theory, these systems can image an entire cleared mouse brain at 500 nm or less resolution without any physical cutting. However, this would require 400+ individual image tiles, and high camera framerates which is problematic for techniques such as axial sweeping. (b-d) By comparison, the ExA-SPIM system uses 1000+ mW lasers, a large-format CMOS camera, and electronics metrology lenses with a moderate NA and ~35 mm working distance. After expanded a mouse brain 3×, the system is still capable of imaging the entire brain in only 15 tiles.

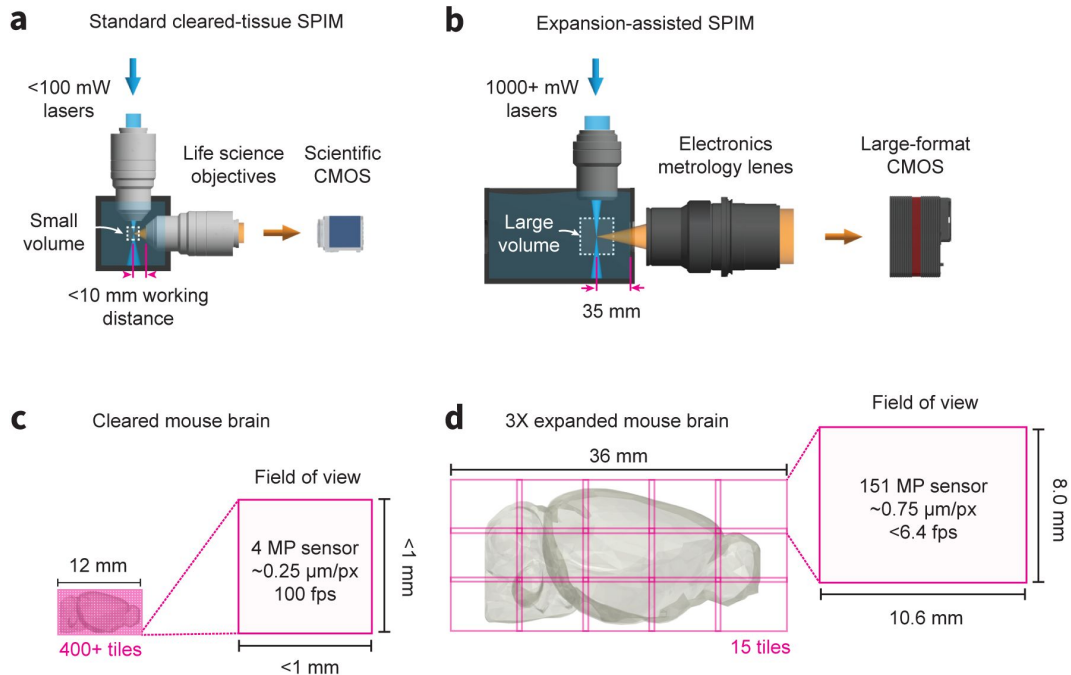


Figure S2

Comparison of electronics metrology and life sciences technologies for large-scale volumetric imaging.

(a) Nikon 20× GLYC next to the VEO_JM DIAMOND 5.0× / F1.3 lens. (b) Traditional sCMOS camera with 2048×2048 pixels next to the VP-151MX camera with the Sony IMX411 sensor with 14192×10640 pixels.

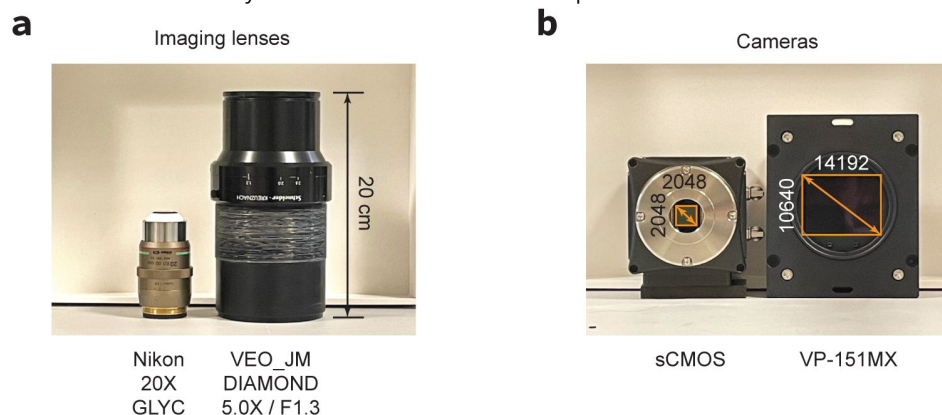


Figure S3

Comparison of methods for large-scale volumetric imaging.

Summary plot comparing the resolution, isotropy, and imaging speed of various existing large volume microscopy methods (2p and SPIM). Methods that do or do not use sectioning are highlighted. The aspect ratio of each marker indicates the relative isotropy of the method. Note that expansion lattice light-sheet microscopy (ExLLSM) (G. Gao et al. 2023) is capable of a very small effective focal volume but over very limited volumes ($\ll 1 \text{ mm}^3$) and is shown in a dedicated inset. The Benchtop mesoSPIM system (Vladimirov et al. 2024), which sits outside of the other SPIM systems is highlighted.

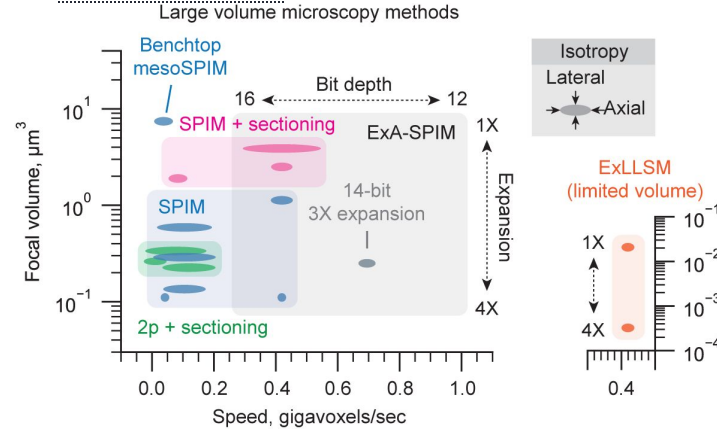
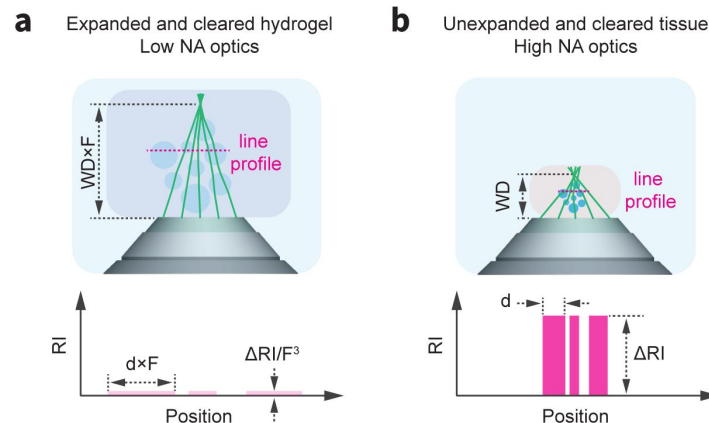


Figure S4

Comparing expansion-assisted versus cleared-tissue imaging.

Although clearing and expansion protocols render tissues transparent, there still exist small refractive index inhomogeneities, ΔRI , which degrade image quality at larger imaging depths. This effect is more extreme for high NA lenses, as light enters the objective over a larger collection angle, leading to path differences between extreme and axial rays. This effect is reduced for low NA optics. For example, spherical aberration (which may be a dominant aberration mode for imaging cleared and expanded tissues) increases with NA to the third power. In other words, tissue-induced aberrations reduce with the expansion factor, F , to the third power. The reduced sensitivity of low NA optics, combined with the increased homogenization of refractive index through the expansion process explain, in part, the benefits of expansion-assisted imaging.



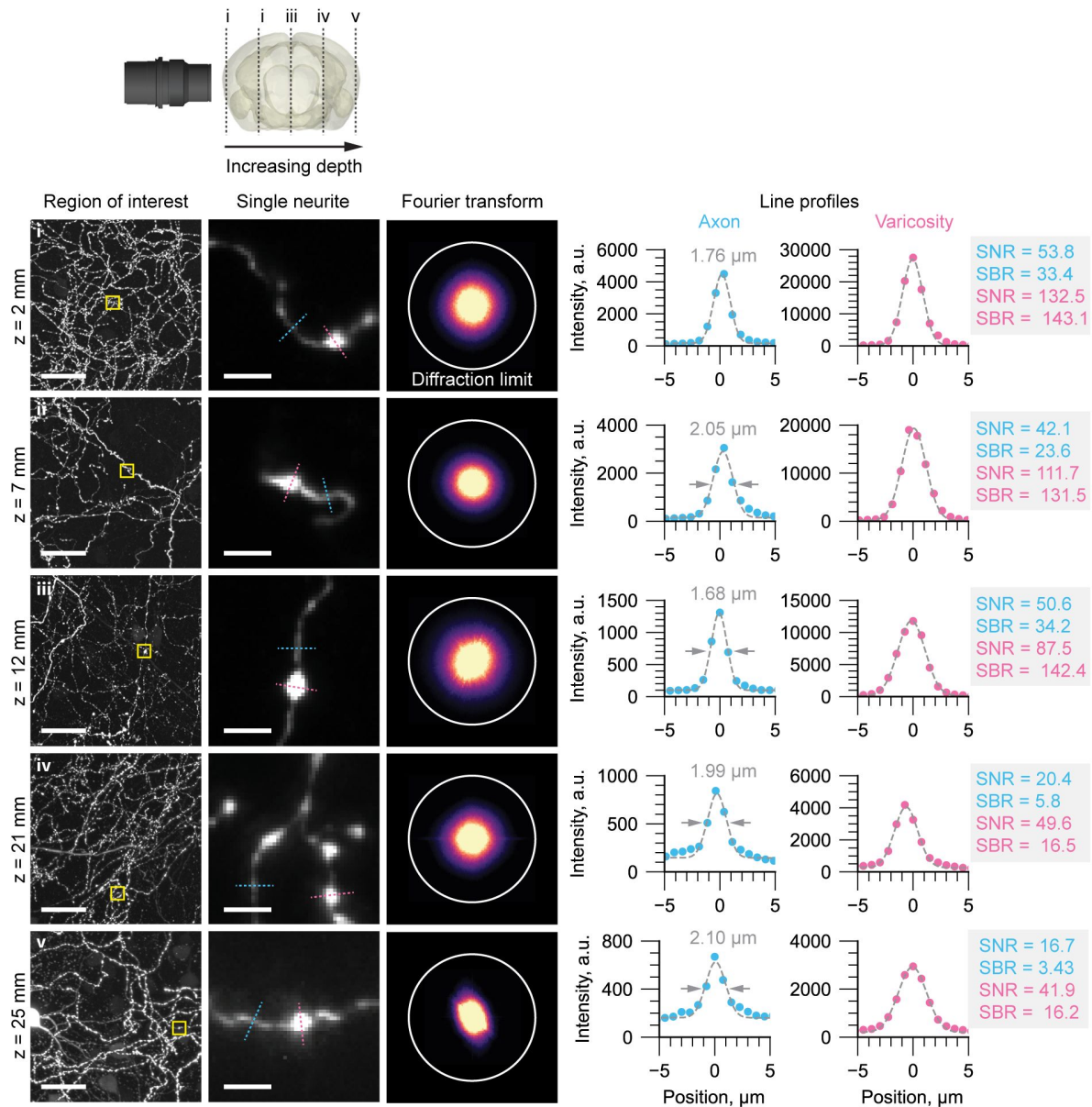


Figure S5

Image quality versus imaging depth.

Representative regions of interest of individual axons in a mouse brain are shown at increasing imaging depths. At each imaging depth, the Fourier transform is shown with a circle denoting the expected diffraction-limited resolution of the microscope. Line profiles through individual axons and varicosities are also shown, with the full-width-half-maximum, signal-to-noise-ratio (SNR), and signal-to-background-ratio (SBR) quantified. The SNR was calculated assuming shot noise limited detection:

Figure S6

Photobleaching at high excitation laser powers.

The same region in an expanded mouse brain sample was irradiated repeatedly to measure photobleaching at the excitation powers used by the ExA-SPIM. Decay curves for a soma (blue) was measured over 2000 repeated exposures. Because the higher excitation power of the ExA-SPIM is distributed over an 11 mm wide light sheet, the light intensity is similar to typical standard SPIM (10^5 - 10^6 mW/cm²). Only modest photobleaching was observed (~50% reduction in intensity after ~800 exposures).

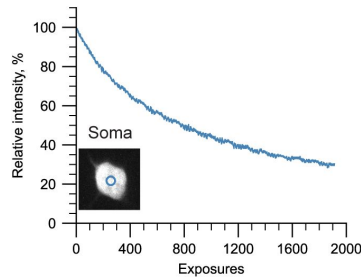


Figure S7

Factors to consider for expansion-assisted imaging.

A continuum of NA and expansion factor combinations can achieve a desired effective resolution. Light collection efficiency (which decreases quadratically with NA) and required working distance (which increases linearly with expansion factor) should be considered when deciding on exact parameters to use for expansion-assisted imaging.

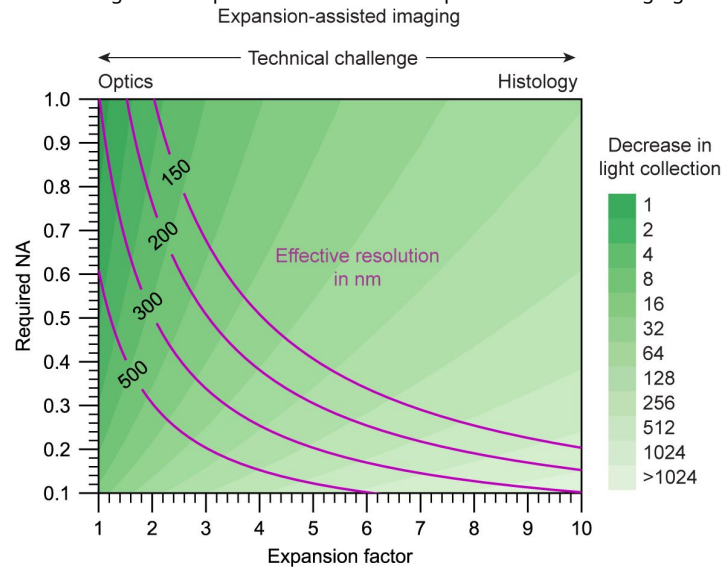


Figure S8

Cleared-tissue imaging with the ExA-SPIM system.

(a) Although we have focused on imaging expanded tissues, the ExA-SPIM microscope does not require expansion. The chamber can be filled with any refractive-index matching media, and the liquid working distance of the electronics metrology lens can be adjusted slightly to recover diffraction-limited imaging performance. (b) As an example, an entire cleared mouse brain could be imaged coronally in a single tile.

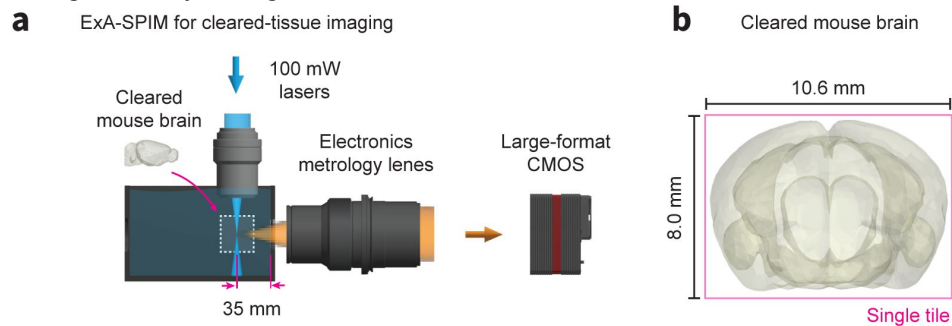
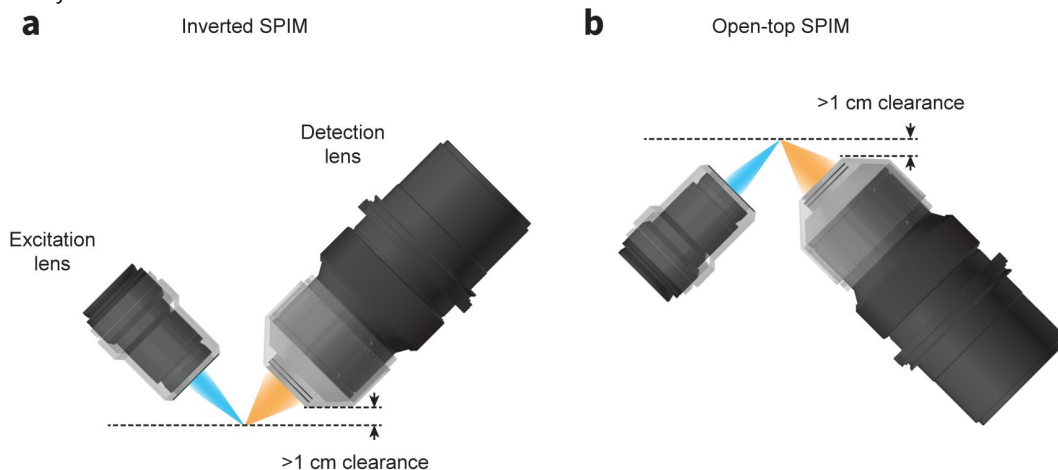


Figure S9

Inverted and open-top ExA-SPIM designs.

Schematics for configuring the ExA-SPIM system in an (a) inverted or (b) open-top architecture. In both geometries, the working distance and mechanical housing of the lenses provides >1 cm of clearance, enabling imaging large tissue sections up to 1 cm thick. This geometry also reduces pathlengths through the tissue, reducing the demands on optical clearing and tissue clarity.



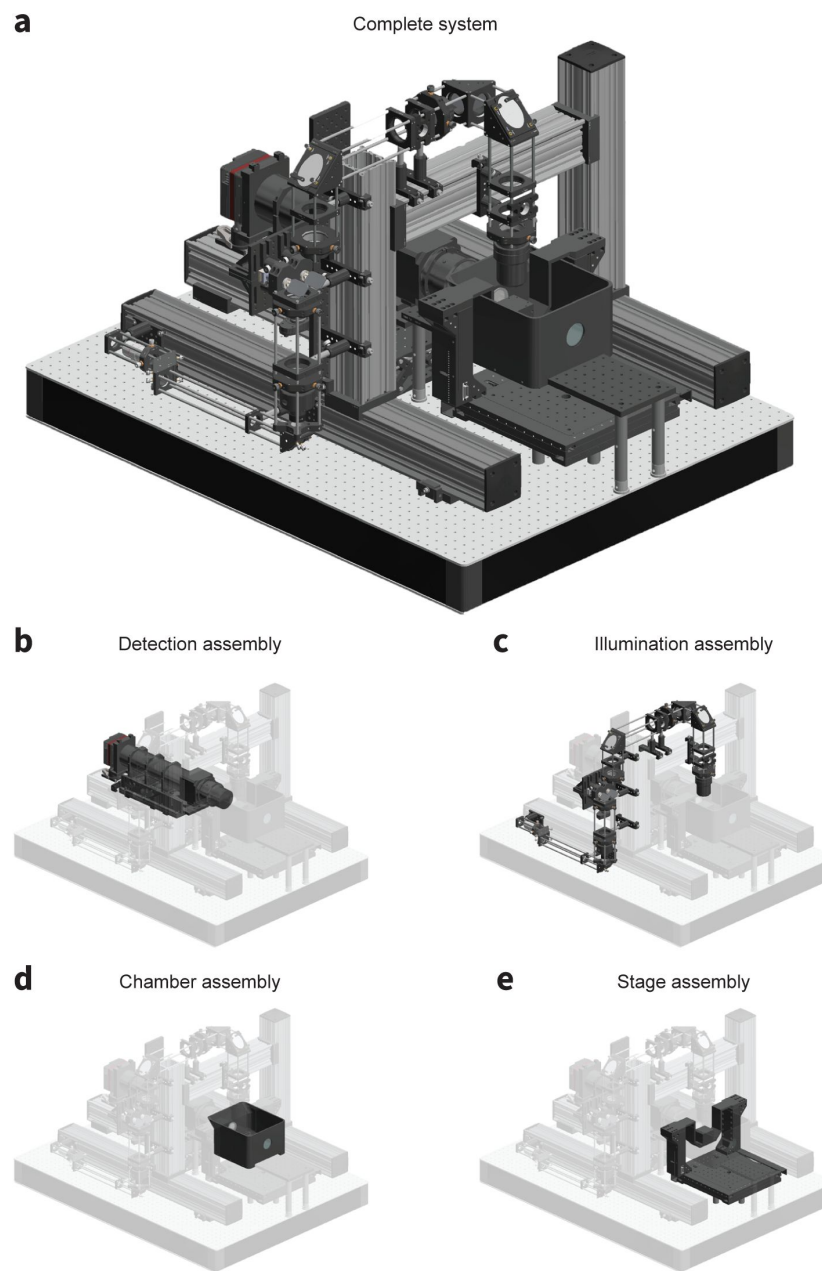


Figure S10

ExA-SPIM mechanical layout.

CAD renderings of the microscope detailing **(a)** the complete system, **(b)** the detection assembly, **(c)** the illumination assembly, **(d)** the chamber assembly, and **(e)** the stage assembly.

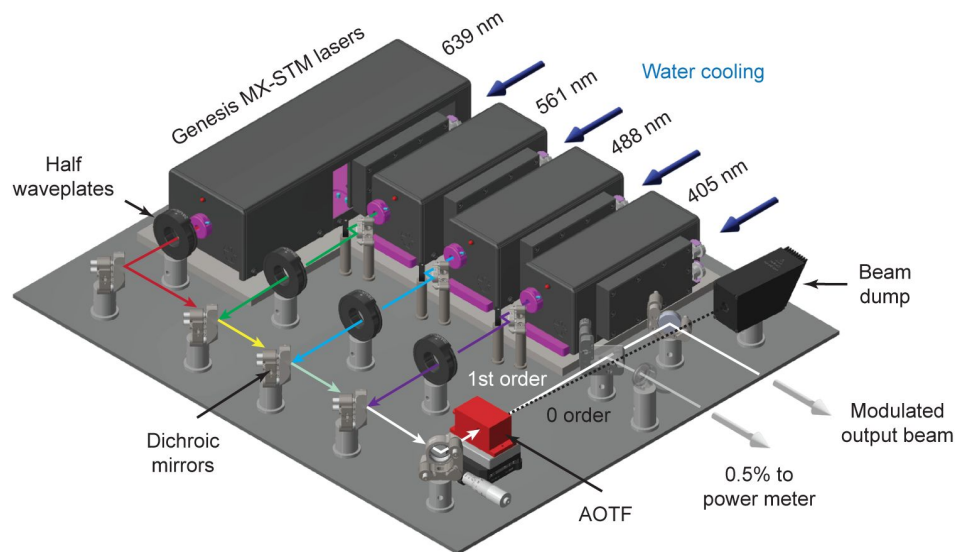


Figure S11

ExA-SPIM laser combiner.

The system uses three 1000 mW lasers at 488, 561, and 639 nm (Genesis MX-STM series, Coherent) with an optional 405 nm laser. The beam from each laser (~ 1 mm in diameter) first passes through a half waveplate for polarization rotation. The beams are all then combined using a series of dichroic mirrors mounted in kinematic mounts. An acousto-optic tunable filter (AOTF) is used to both select wavelength and modulate the power of each laser. The 0-order beam from the AOTF terminates in a beam dump, whereas the modulated 1st order beam is reflected off a final kinematic mirror before being injected into the microscope. An AR coated glass plate is used to reflect $\sim 0.5\%$ of the output beam to a power meter for monitoring during dataset acquisition.

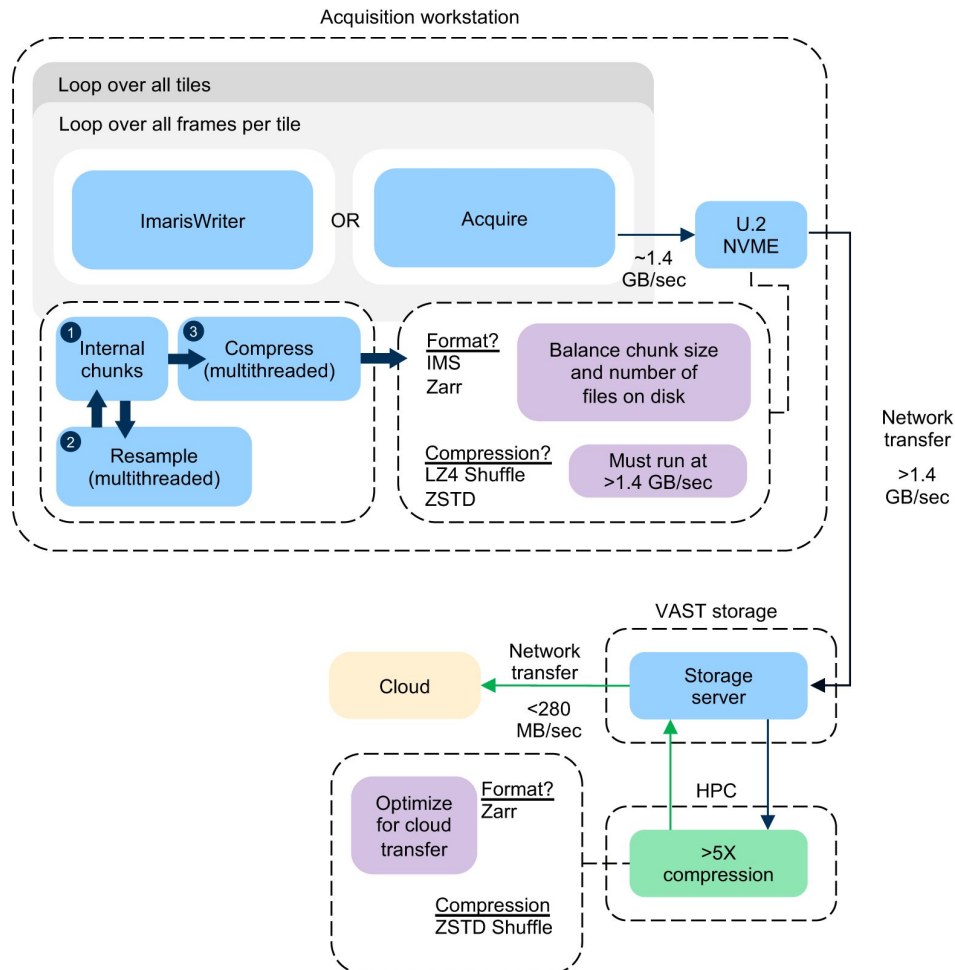


Figure S12

ExA-SPIM image acquisition scheme.

A block diagram summarizing the acquisition procedure for an ExA-SPIM dataset is shown. Each dataset requires looping over the total frames within a given tile, and then all tiles within a given dataset. Each tile results in a single file on disk, which is copied over the network to a local VAST storage system. The transfer of the previous tile occurs synchronously with the acquisition of the next tile, and the transfer speed to the VAST system outpaces the data generation speed of the microscope. For the ImarisWriter workflow, once tiles are transferred to the VAST storage system, they are synchronously converted to OME-Zarr files with ZSTD Shuffle compression. After compression, the files are written directly to cloud storage. The compression and conversion to OME-Zarr also runs at a speed which outpaces the microscope.

Figure S13

Simulations of camera sensitivity.

(a) Simulations of SNR versus collected photons for sCMOS (cyan) and the Sony IMX411 sensor with 16-bit (green), 14-bit (yellow), and 12-bit (red) readout. The SNR for an ideal perfect sensor is highlighted. (b) Relative SNR values for the curves shown in (a), normalized to the ideal sensor curve.

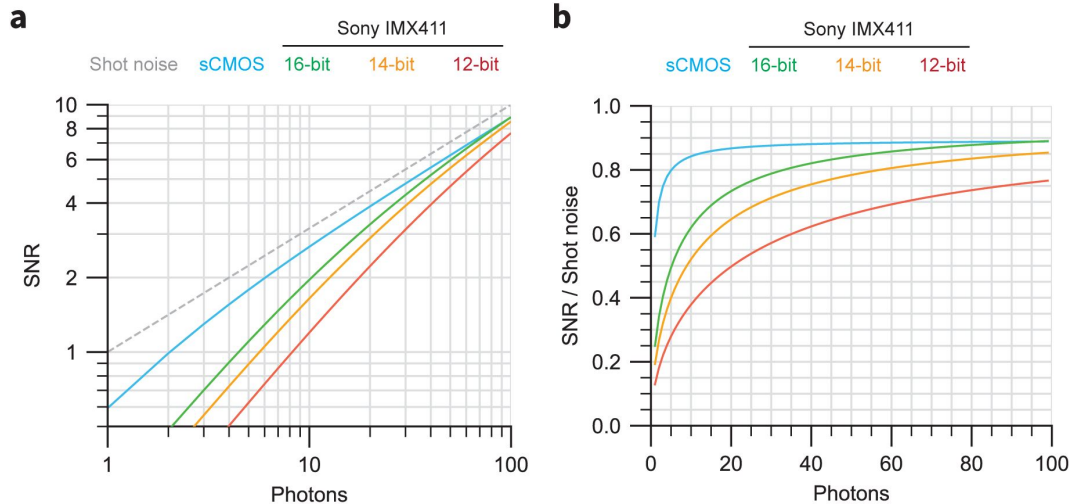


Table S1

Summary of custom lenses for fluorescence microscopy.

Reference	Name	Liquid immersion	f	NA	Field of view	Etendue	Working distance
(Tang et al. 2024)	Curved LSFM	1.33 - 1.56	118	0.25	13.00	8.30	20.00
(Current study)	ExA-SPIM	1.33 - 1.56	100	0.31	16.40	19.65	40.00
(Fan et al. 2019)	RUSH	-	-	0.35	14.00	18.86	19.00
(Keisuke Ota et al. 2020)	FASHIO-2PM	-	35	0.40	4.24	2.26	4.50
(McConnell et al. 2016)	Mesolens	1.33 - 1.56	50	0.47	6.00	1.13	3.00
(Yu et al. 2024)	Cousa	-	20	0.50	2.00	0.79	20.00
(Yu et al. 2021)	Diesel2P	-	30	0.54	6.00	8.24	8.00
(Kyocera)	CS06-10-40-154	1.52 - 1.56	18	0.60	2.00	1.13	40.60
(Sofroniew et al. 2016)	2p-RAM	-	21	0.60	5.00	7.07	2.70
(Voigt et al. 2024)	Schmidt	1.33 - 1.56	-	1.00	1.15	1.04	11.00

Units
 f - mm
field of view - mm
etendue - mm²
working distance - mm

Reference	Imaging method	Optical resolution	Voxel size	Isotropy	Focal volume	Voxel rate
(Economo et al. 2016)	2p tomography	0.45×0.45×1.33	0.30×0.30×1.00	2.96:1	0.27	16
(Gong et al. 2016)	fMOST	0.32×0.32×2.00	0.32×0.32×2.00	6.25:1	0.20	4
(Narasimhan et al. 2017)	Oblique light-sheet	0.75×0.75×6.90	0.41×0.41×0.41	9.20:1	3.88	419
(Migliori et al. 2018)	Light-sheet theta	0.34×0.34×3.00	0.23×0.23×5.00	8.82:1	0.29	105
(Chakraborty et al. 2019)	Axially-swept SPIM	0.48×0.48×0.48	0.16×0.16×0.16	1:1	0.11	42
(Voleti et al. 2019)	SCAPE 2.0	0.60×1.21×1.55	0.24×0.24×0.24	2.58:1	1.13	419
(R. Gao et al. 2019)	ExLLSM	0.06×0.06×0.09	0.03×0.03×0.04	1.5:1	«0.01	68
(D. Chen et al. 2020)	Multifocal 2p tomography	0.36×0.36×2.59	0.40×0.40×1.00	7.19:1	0.34	77
(Guo et al. 2020)	Dual-inverted SPIM	0.48×0.48×0.48	0.24×0.24×0.24	1:1	0.11	419
(Y. Chen et al. 2020)	Tiling SPIM	0.30×0.30×2.50	0.30×0.30×1.50	5:1	0.22	105
(Wang et al. 2021)	fMOST	0.34×0.34×2.00	0.23×0.23×1.00	5.88:1	0.23	136
(Zhang et al. 2021)	Axially-swept SPIM	0.95×0.95×2.10	0.52×0.52×1.00	2.21:1	1.90	84
(Xu et al. 2021)	VISOR2 SPIM	1.00×1.00×2.50	1.00×1.00×2.50	2.5:1	2.50	373
(Glaser et al. 2022)	Open-top light-sheet	0.45×0.45×2.91	0.20×0.20×0.20	6.47:1	0.59	105
(Xiaoli Qi et al. 2023)	Confocal airy light-sheet	0.76×0.76×2.20	0.26×0.26×1.06	2.08:1	1.27	419
(Vladimirov et al. 2024)	Benchtop mesoSPIM	1.50×1.50×3.30	0.75×0.75×2.00	2.2:1	7.43	37
(Tang et al. 2024)	Curved LSFM	1.22×1.22×2.50	0.63×0.63×1.25	2:1	3.72	130
(Current study)	ExA-SPIM (1×)	1.00×1.00×3.00	0.75×0.75×1.00	3:1	6.75	725
(Current study)	ExA-SPIM (2×)	0.50×0.50×1.50	0.38×0.38×0.50	3:1	0.84	725
(Current study)	ExA-SPIM (3×)	0.33×0.33×1.00	0.25×0.25×0.33	3:1	0.25	725
(Current study)	ExA-SPIM (4×)	0.25×0.25×0.75	0.19×0.19×0.25	3:1	0.11	725

Units

optical resolution - μm

voxel size - μm

focal volume - μm^3

voxel rate - megavoxels/sec

Note, only modalities capable of providing $\leq 1.5 \mu\text{m}$ lateral optical resolution are included.

Table S2

Comparison of large-scale volumetric imaging modalities.

Manufacturer	Model	Magnification	NA	Lateral resolution	Field of view	Etendue	Speed
Schneider-Kreuznach	VEO_JM DIAMOND 1.43×/F3.0	1.43	0.10	3.05	57.34	26.34	2137
Schneider-Kreuznach	VEO_JM DIAMOND 1.67×/F3.0	1.67	0.10	3.05	49.10	19.32	1340
Schneider-Kreuznach	VEO_JM DIAMOND 2.5×/F2.6	2.50	0.13	2.35	32.80	15.17	400
Schneider-Kreuznach	VEO_JM DIAMOND 3.33×/F2.1	3.33	0.18	1.69	24.62	15.43	170
Schneider-Kreuznach	VEO_JM DIAMOND 5.0×/F1.3	5.00	0.31	1.00	16.40	19.65	50
Nikon	Rayfact 1.4S	1.40	0.11	2.77	61.71	36.19	2277
Nikon	Rayfact 1.7S	1.70	0.11	2.77	49.37	23.16	1272
Nikon	Rayfact 2.5S	2.50	0.14	2.18	34.56	18.39	400
Nikon	Rayfact 3.5S	3.50	0.16	1.91	24.69	12.25	146
Nikon	Rayfact 5S	5.00	0.17	1.79	17.28	6.78	50
Nikon	Rayfact 1-2x Variable Lens	1.00	0.09	3.39	86.4	47.49	6249
Nikon	Rayfact 1-2x Variable Lens	2.00	0.12	2.54	43.2	21.11	781
Nikon	Rayfact 2-5x Variable Lens	2.00	0.13	2.35	43.2	24.77	781
Nikon	Rayfact 2-5x Variable Lens	5.00	0.17	1.79	16.6	6.25	50

Units
lateral resolution - μm
field of view - mm
etendue - mm^2
speed - cm^3 per day

Table S3

ExA-SPIM imaging across scales.

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Reviewer #1 (Public Review):

Summary:

Glaser et al present ExA-SPIM, a light-sheet microscope platform with large volumetric coverage (Field of view 85mm^2 , working distance 35mm), designed to image expanded mouse brains in their entirety. The authors also present an expansion method optimized for whole mouse brains, and an acquisition software suite. The microscope is employed in imaging an expanded mouse brain, the macaque motor cortex and human brain slices of white matter.

This is impressive work, and represents a leap over existing light-sheet microscopes. As an example, it offers a ~ fivefold higher resolution than mesoSPIM (<https://mesospim.org/>), a popular platform for imaging large cleared samples. Thus while this work is rooted in optical engineering, it manifests a huge step forward and has the potential to become an important tool in the neurosciences.

Strengths:

- ExA-SPIM features an exceptional combination of field of view, working distance, resolution and throughput.

- An expanded mouse brain can be acquired with only 15 tiles, lowering the burden on computational stitching. That the brain does not need to be mechanically sectioned is also seen as an important capability.

- The image data is compelling, and tracing of neurons has been performed. This demonstrates the potential of the microscope platform.

Weaknesses:

- There is a general question about the scaling laws of lenses, and expansion microscopy, which in my opinion remained unanswered: In the context of whole brain imaging, a larger expansion factor requires a microscope system with larger volumetric coverage, which in turn will have lower resolution (Figure 1B). So what is optimal? Could one alternatively image a cleared (non-expanded) brain with a high resolution ASLM system (Chakraborty, Tonmoy, Nature Methods 2019, potentially upgraded with custom objectives) and get similar effective resolution as the authors get with expansion? This is not meant to diminish the achievement, but it was unclear if the gains in resolution from the expansion factor are traded off by the scaling laws of current optical systems.

- It was unclear if 300 nm lateral and 800 nm axial resolution is enough for many questions in neuroscience. Segmenting spines, distinguishing pre- and postsynaptic densities, or tracing densely labeled neurons might be challenging. A discussion about the necessary resolution levels in neuroscience would be appreciated.

- Would it be possible to characterize the aberrations that might be still present after whole brain expansion? One approach could be to image small fluorescent nanospheres behind the expanded brain, and recover the pupil function via phase retrieval. But even full width half maximum (FWHM) measurements of the nanospheres' images would give some idea of the magnitude of the aberrations.

Review of the revised manuscript:

The authors have carefully addressed my concerns and suggestions.

I appreciate the extended discussion on tissue clearing compared to expansion. I would recommend substantiating some of the statements though with references, or in other instances expanding a little further. I would encourage the authors to consider the points below. But there is also another path to actually reduce that specific discussion, if the conclusion is that it opened more questions than answers.

Specifically, here are some points in the paragraph that discusses tissue clearing and expansion that could be improved:

- The statement "Spherical aberration increases with NA" reads nonspecific to me. I think a more precise formulation would be "The effect of spherical aberration (e.g. loss of Strehl ratio) increases with NA. The stated third power law would also benefit from a reference.

- The statement "the index of refraction gradients in tissue decreases with the third power of the expansion factor..." reads a bit odd. "Gradients in refractive index" would be more consistent with the usage of r.i. throughout the manuscript.

For the third power law, it might be important to know what drives the remaining refractive index variation in expansion microscopy. If it is the labels and their linkers, then indeed, they get increasingly diluted as their amount remains constant. However, if the aberrations are caused by the polymer gel, I would assume you would need more monomer material for higher expansion factors? Thus, I was not fully sure about the scaling law in this case. If there is a reference where this was explored in detail, that would resolve this issue.

- The statement that aberrations scale with gradients in refractive index also needs either a reference, or an explanation for the reader. I think figure S4 was supposed to illustrate this, but was not referenced in the discussion (and could be clarified, see comment below).

To me, the discussion focused strongly on tissue clearing vs expansion. What was left out in the discussion was if larger expansion factors would be favorable (i.e. whole brain imaging with 10-20X expansion instead of 4-5X). Some arguments implicitly seemed to stipulate that a larger expansion factor would optically be favorable. But Figure S7 highlights another tradeoff with the decay in sensitivity and Figure 1b provides the technological constraints on lens design. So as a reader, I was not fully sure if the next frontier should be 10-20X expansion brain imaging, or if 4-5X is currently a sweet spot.

Further comments:

Please explain the variables in Figure S4, such as F, WD and d. It was unclear to me what the RI profile should mean in the bottom row. Naively, the figure of merit would be the optical path length that is integrated along the different rays, as this leads to a variation in the wavefront.

Figure S5: I would caution to say the SNR was quantified, but rather say it was estimated (in the shot noise limit). Was the background subtracted for the SNR measurements?

Squaring the SNR estimates, it looks like the photon counts went down ~10-fold from $z=2\text{mm}$ to $z=25\text{mm}$. That is a larger reduction in signal than I had expected. If it was based solely on aberrations, a 10-fold drop in Strehl ratio seems significant (potentially smaller if we assume the light-sheet also underwent aberrations). Are there other factors that could explain the signal reduction (maybe from the labeling side)?

Further on Figure S5: Fourier transforms (power spectrum) and single line profiles are in my opinion not the best way to quantify resolution. Could the authors perform image decorrelation analysis on the region of interest (Descloux, A., Kristin Stefanie Großmayer, and Aleksandra Radenovic. "Parameter-free image resolution estimation based on decorrelation analysis." *Nature methods* 2019) or Fourier ring correlation? This would give in some sense an average resolving power in that depth, and would remove the bias from picking a line profile.

<https://doi.org/10.7554/eLife.91979.2.sa2>

Reviewer #2 (Public Review):

Summary:

In this revised manuscript, Glaser et al. have responded to the reviewer comments by removing some of the overstated claims from the prior manuscript and editing portions of the manuscript text to enhance the clarity. Although the manuscript would be stronger if the authors had been able to provide data that justified the original high-impact claims from the initial publication (e.g. that the images could be used for robust and automated neuronal tracing across large volumes), the amended manuscript text now more closely matches the supporting data. As with the initial submission, I believe that the microscope design and characterization is a useful contribution to the field and the data are quite stunning. However, I still feel like there are some overstated claims in this revision that should be addressed so as not to mislead readers.

<https://doi.org/10.7554/eLife.91979.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

Glaser et al present ExA-SPIM, a light-sheet microscope platform with large volumetric coverage (Field of view 85mm^2 , working distance 35mm), designed to image expanded mouse brains in their entirety. The authors also present an expansion method optimized for whole mouse brains and an acquisition software suite. The microscope is employed in imaging an expanded mouse brain, the macaque motor cortex, and human brain slices of white matter.

This is impressive work and represents a leap over existing light-sheet microscopes. As an example, it offers a fivefold higher resolution than mesoSPIM (<https://mesospim.org/>), a popular platform for imaging large cleared samples. Thus while this work is rooted in optical engineering, it manifests a huge step forward and has the potential to become an important tool in the neurosciences.

Strengths:

- ExA-SPIM features an exceptional combination of field of view, working distance, resolution, and throughput.
- An expanded mouse brain can be acquired with only 15 tiles, lowering the burden on computational stitching. That the brain does not need to be mechanically sectioned is also seen as an important capability.
- The image data is compelling, and tracing of neurons has been performed. This demonstrates the potential of the microscope platform.

Weaknesses:

- There is a general question about the scaling laws of lenses, and expansion microscopy, which in my opinion remained unanswered: In the context of whole brain imaging, a larger expansion factor requires a microscope system with larger volumetric coverage, which in turn will have lower resolution (Figure 1B). So what is optimal? Could one alternatively image a cleared (non-expanded) brain with a high-resolution ASLM system (Chakraborty, Tonmoy, Nature Methods 2019, potentially upgraded with custom

objectives) and get a similar effective resolution as the authors get with expansion? This is not meant to diminish the achievement, but it was unclear if the gains in resolution from the expansion factor are traded off by the scaling laws of current optical systems.

Paraphrasing the reviewer: Expanding the tissue requires imaging larger volumes and allows lower optical resolution. What has been gained?

The answer to the reviewer's question is nuanced and contains four parts.

First, optical engineering requirements are more forgiving for lenses with lower resolution. Lower resolution lenses can have much larger fields of view (in real terms: the number of resolvable elements, proportional to 'etendue') and much longer working distances. In other words, it is currently more feasible to engineer lower resolution lenses with larger volumetric coverage, even when accounting for the expansion factor.

Second, these lenses are also much better corrected compared to higher resolution (NA) lenses. They have a flat field of view, negligible pincushion distortions, and constant resolution across the field of view. We are not aware of comparable performance for high NA objectives, even when correcting for expansion.

Third, although clearing and expansion render tissues 'transparent', there still exist refractive index inhomogeneities which deteriorate image quality, especially at larger imaging depths. These effects are more severe for higher optical resolutions (NA), because the rays entering the objective at higher angles have longer paths in the tissue and will see more aberrations. For lower NA systems, such as ExaSPIM, the differences in paths between the extreme and axial rays are relatively small and image formation is less sensitive to aberrations.

Fourth, aberrations are proportional to the index of refraction inhomogeneities (dn/dx). Since the index of refraction is roughly proportional to density, scattering and aberration of light decreases as M^3 , where M is the expansion factor. In contrast, the imaging path length through the tissue only increases as M . This produces a huge win for imaging larger samples with lower resolutions.

To our knowledge there are no convincing demonstrations in the literature of diffraction-limited ASLM imaging at a depth of 1 cm in cleared mouse brain tissue, which would be equivalent to the ExA-SPIM imaging results presented in this manuscript.

In the discussion of the revised manuscript we discuss these factors in more depth.

- It was unclear if 300 nm lateral and 800 nm axial resolution is enough for many questions in neuroscience. Segmenting spines, distinguishing pre- and postsynaptic densities, or tracing densely labeled neurons might be challenging. A discussion about the necessary resolution levels in neuroscience would be appreciated.

We have previously shown good results in tracing the thinnest (100 nm thick) axons over cm scales with 1.5 μ m axial resolution. It is the contrast (SNR) that matters, and the ExaSPIM contrast exceeds the block-face 2-photon contrast, not to mention imaging speed ($> 10\times$).

Indeed, for some questions, like distinguishing fluorescence in pre- and postsynaptic structures, higher resolutions will be required (0.2 μ m isotropic; Rah et al Frontiers Neurosci, 2013). This could be achieved with higher expansion factors.

This is not within the intended scope of the current manuscript. As mentioned in the discussion section, we are working towards ExA-SPIM-based concepts to achieve better resolution through the design and fabrication of a customized imaging lens that maintains a high volumetric coverage with increased numerical aperture.

- Would it be possible to characterize the aberrations that might be still present after whole brain expansion? One approach could be to image small fluorescent nanospheres behind the expanded brain and recover the pupil function via phase retrieval. But even full width half maximum (FWHM) measurements of the nanospheres' images would give some idea of the magnitude of the aberrations.

We now included a supplementary figure highlighting images of small axon segments within distal regions of the brain.

Reviewer #2 (Public Review):

Summary:

In this manuscript, Glaser et al. describe a new selective plane illumination microscope designed to image a large field of view that is optimized for expanded and cleared tissue samples. For the most part, the microscope design follows a standard formula that is common among many systems (e.g. Keller PJ et al Science 2008, Pitrone PG et al. Nature Methods 2013, Dean KM et al. Biophys J 2015, and Voigt FF et al. Nature Methods 2019). The primary conceptual and technical novelty is to use a detection objective from the metrology industry that has a large field of view and a large area camera. The authors characterize the system resolution, field curvature, and chromatic focal shift by measuring fluorescent beads in a hydrogel and then show example images of expanded samples from mouse, macaque, and human brain tissue.

Strengths:

I commend the authors for making all of the documentation, models, and acquisition software openly accessible and believe that this will help assist others who would like to replicate the instrument. I anticipate that the protocols for imaging large expanded tissues (such as an entire mouse brain) will also be useful to the community.

Weaknesses:

The characterization of the instrument needs to be improved to validate the claims. If the manuscript claims that the instrument allows for robust automated neuronal tracing, then this should be included in the data.

The reviewer raises a valid concern. Our assertion that the resolution and contrast is sufficient for robust automated neuronal tracing is overstated based on the data in the paper. We are hard at work on automated tracing of datasets from the ExA-SPIM microscope. We have demonstrated full reconstruction of axonal arbors encompassing >20 cm of axonal length. But including these methods and results is out of the scope of the current manuscript.

The claims of robust automated neuronal tracing have been appropriately modified.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Smaller questions to the authors:

- Would a multi-directional illumination and detection architecture help? Was there a particular reason the authors did not go that route?

Despite the clarity of the expanded tissue, and the lower numerical aperture of the ExA-SPIM microscope, image quality still degrades slightly towards the distal regions of the brain

relative to both the excitation and detection objective. Therefore, multi-directional illumination and detection would be advantageous. Since the initial submission of the manuscript, we have undertaken re-designing the optics and mechanics of the system. This includes provisions for multi-directional illumination and detection. However, this new design is beyond the scope of this manuscript. We now mention this in L254-255 of the Discussion section.

- Why did the authors not use the same objective for illumination and detection, which would allow isotropic resolution in ASLM?

The current implementation of ASLM requires an infinity corrected objective (i.e. conjugating the axial sweeping mechanism to the back focal plane). This is not possible due to the finite conjugate design of the ExA-SPIM detection lens.

More fundamentally, pushing the excitation NA higher would result in a shorter light sheet Rayleigh length, which would require a smaller detection slit (shorter exposure time, lower signal to noise ratio). For our purposes an excitation NA of 0.1 is an excellent compromise between axial resolution, signal to noise ratio, and imaging speed.

For other potentially brighter biological structures, it may be possible to design a custom infinity corrected objective that enables ASLM with NA > 0.1.

- Have the authors made any attempt to characterize distortions of the brain tissue that can occur due to expansion?

We have not systematically characterized the distortions of the brain tissue pre and post expansion. Imaged mouse brain volumes are registered to the Allen CCF regardless of whether or not the tissue was expanded. It is beyond the scope of this manuscript to include these results and processing methods, but we have confirmed that the ExA-SPIM mouse brain volumes contain only modest deformation that is easily accounted for during registration to the Allen CCF.

- The authors state that a custom lens with NA 0.5-0.6 lens can be designed, featuring similar specifications. Is there a practical design? Wouldn't such a lens be more prone to field curvature?

This custom lens has already been designed and is currently being fabricated. The lens maintains a similar space bandwidth product as the current lens (increased numerical aperture but over a proportionally smaller field of view). Over the designed field of view, field curvature is <1 μm . However, including additional discussion or results of this customized lens is beyond the scope of this manuscript.

Reviewer #2 (Recommendations For The Authors):

• *System characterization:*

- Please state what wavelength was used for the resolution measurements in Figure 2.

An excitation wavelength of 561 nm was used. This has been added to the manuscript text.

- The manuscript highlights that a key advance for the microscope is the ability to image over a very large 13 mm diameter field of view. Can the authors clarify why they chose to characterize resolution over an 8mm diameter field rather than the full area?

The 13 mm diameter field of view refers to the diagonal of the 10.6 x 8.0 mm field of view. The results presented in Figure 1c are with respect to the horizontal x direction and vertical y direction. A note indicating that the 13 mm is with respect to the diagonal of the rectangular imaging field has been added to the manuscript text. The results were presented in this way to present the axial and lateral resolution as a function of y (the axial sweeping direction).

- The resolution estimates seem lower than I would expect for a 0.30 NA lens (which should be closer to ~850 nm for 515 nm emission). Could the authors clarify the discrepancy? Is this predicted by the Zemax model and due to using the lens in immersion media, related to sampling size on the camera, or something else? It would be helpful if the authors could overlay the expected diffraction-limited performance together with the plots in Figure 2C.

As mentioned previously, the resolution measurements were performed with 561 nm excitation and an emission bandpass of ~573 – 616 nm (595 nm average). Based on this we would expect the full width half maximum resolution to be ~975 nm. The resolution is in fact limited by sampling on the camera. The 3.76 μm pixel size, combined with the 5.0X magnification results in a sampling of 752 nm. Based on the Nyquist the resolution is limited to ~1.5 μm . We have added clarifying statements to the text.

- I'm confused about the characterization of light sheet thickness and how it relates to the measured detection field curvature. The authors state that they "deliver a light sheet with NA = 0.10 which has a width of 12.5 mm (FWHM)." If we estimate that light fills the 0.10 NA, it should have a beam waist (2wo) of ~3 microns (assuming Gaussian beam approximations). Although field curvature is described as "minimal" in the text, it is still ~10-15 microns at the edge of the field for the emission bands for GFP and RFP proteins. Given that this is 5X larger than the light sheet thickness, how do the authors deal with this?

The generated light sheet is flat, with a thickness of ~ 3 μm . This flat light sheet will be captured in focus over the depth of focus of the detection objective. The stated field curvature is within 2.5X the depth of focus of the detection lens, which is equivalent to the "Plan" specification of standard microscope objectives.

- In Figure 2E, it would be helpful if the authors could list the exposure times as well as the total voxels/second for the two-camera comparison. It's also worth noting that the Sony chip used in the VP151MX camera was released last year whereas the Orca Flash V3 chosen for comparison is over a decade old now. I'm confused as to why the authors chose this camera for comparison when they appear to have a more recent Orca BT-Fusion that they show in a picture in the supplement (indicated as Figure S2 in the text, but I believe this is a typo and should be Figure S3).

This is a useful addition, and we have added exposure times to the plot. We have also added a note that the Orca Flash V3 is an older generation sCMOS camera and that newer variants exist. Including the Orca BT-Fusion. The BT-Fusion has a read noise of 1.0 e- rms versus 1.6 e- rms, and a peak quantum efficiency of ~95% vs. 85%. Based on the discussion in Supplementary Note S1, we do not expect that these differences in specifications would dramatically change the data presented in the plot. In addition, the typo in Figure S2 has been corrected to Figure S3.

- In Table S1, the authors note that they only compare their work to prior modalities that are capable of providing ≤ 1 micron resolution. I'm a bit confused by this choice given that Figure 2 seems to show the resolution of ExA-SPIM as ~1.5 microns at 4 mm off

center (1/2 their stated radial field of view). It also excludes a comparison with the mesoSPIM project which at least to me seems to be the most relevant prior to this manuscript. This system is designed for imaging large cleared tissues like the ones shown here. While the original publication in 2019 had a substantially lower lateral resolution, a newer variant, Nikita et al bioRxiv (which is cited in general terms in this manuscript, but not explicitly discussed) also provides 1.5-micron lateral resolution over a comparable field of view.

We have updated the table to include the benchtop mesoSPIM from Nikita et al., Nature Communications, 2024. Based on this published version of the manuscript, the lateral resolution is 1.5 μm and axial resolution is 3.3 μm . Assuming the Iris 15 camera sensor, with the stated 2.5 fps, the volumetric rate (megavoxels/sec) is 37.41.

- The authors state that, "We systematically evaluated dehydration agents, including methanol, ethanol, and tetrahydrofuran (THF), followed by delipidation with commonly used protocols on 1 mm thick brain slices. Slices were expanded and examined for clarity under a microscope." It would be useful to include some data from this evaluation in the manuscript to make it clear how the authors arrived at their final protocol.

Additional details on the expansion protocol may be included in another manuscript.

General comments:

- There is a tendency in the manuscript to use negative qualitative terms when describing prior work and positive qualitative terms when describing the work here. Examples include:
 - "Throughput is limited in part by cumbersome and error-prone microscopy methods". While I agree that performing single neuron reconstructions at a large scale is a difficult challenge, the terms cumbersome and error-prone are qualitative and lacking objective metrics.

We have revised this statement to be more precise, stating that throughput is limited in part by the speed and image quality of existing microscopy methods.

- The resolution of the system is described in several places as "near-isotropic" whereas prior methods were described as "highly anisotropic". I agree that the ~1:3 lateral to axial ratio here is more isotropic than the 1:6 ratio of the other cited publications. However, I'm not sure I'd consider 3-fold worse axial resolution than lateral to be considered "near" isotropic.

We agree that the term near-isotropic is ambiguous. We have modified the text accordingly, removing the term near-isotropic and where appropriate stating that the resolution is more isotropic than that of other cited publications.

- exposures (which in the caption is described as "modest"). I'd suggest removing these qualitative terms and just stating the values.

We agree and have changed the text accordingly.

- The results section for Figure 5 is titled "Tracing axons in human neocortex and white matter". Although this section states "larger axons (>1 μm) are well separated... allowing for robust automated and manual tracing" there is no data for any tracing in the manuscript. Although I agree that the images are visually impressive, I'm not sure that this claim is backed by data.

We have now removed the text in this section referring to automated and manual tracing.

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